

Domestic Wedges and the (In)Sensitivity of CPI to Exchange Rates

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Abstract

Earlier literature highlights domestic wedges—distribution costs, variable markups, and nominal rigidities—in limiting exchange rate pass-through to consumer prices. Using granular Chilean data, I quantify their impact, considering input-output linkages and sectoral heterogeneity. Domestic wedges reduce CPI sensitivity to exchange rate fluctuations by 57%. Distribution costs account for 48% of the overall reduction in sensitivity, variable markups for 29%, and nominal rigidities for 23%. These factors dampen CPI sensitivity more than incomplete pass-through into prices at the border. Eliminating sectoral heterogeneity in wedges increases CPI response by 55%, as import exposure positively correlates with wedge strength. Ignoring this heterogeneity misidentifies key sectors driving exchange rate effects on consumer prices. Contrary to earlier quantification strategies, I show that CPI sensitivity mainly stems from imported consumption goods, as domestic wedges weaken the amplification role of input-output linkages.

Keywords: Exchange Rates, Pass-Through, Input-Output Linkages, Wedges, Variable Markups.
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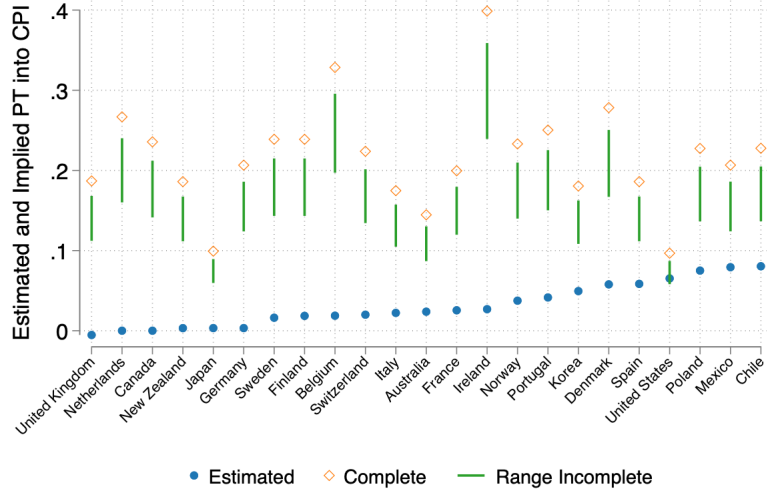
A well-documented stylized fact in international economics—one with far-reaching implications, from the design of optimal monetary policy in open economies to the dynamics of domestic inequality—is the low sensitivity of inflation to exchange rate fluctuations.¹ The blue dots in Figure 1 illustrates this pattern for OECD countries over 1990–2019: using standard econometric strategies, the estimated sensitivity of CPI to a one percent exchange rate change is typically below 0.1, with an average of 0.032 across countries. A natural explanation is that inflation is only weakly exposed to exchange rates because imported goods constitute a limited share of consumption. Yet, when we compute the CPI sensitivity implied by the import share in consumption (orange squares), we find that the estimated sensitivity is on average four to six times lower than predicted. This discrepancy persists even after accounting for incomplete pass-through of exchange rate changes into import prices. This paper presents extensive quantitative evidence showing that the muted response of domestic prices is due to the presence of wedges—namely, (a) pass-through at the border, (b) distribution costs, (c) markup variation, and (d) nominal price rigidities—with (b) through (d) corresponding to domestic wedges.

I develop an analytical framework to quantify the expected sensitivity of the CPI to exchange rate fluctuations, accounting for domestic wedges and incomplete pass-through at the border (Goldberg and Campa, 2010; Nakamura and Zerom, 2010). CPI sensitivity arises through three channels: direct exposure via consumption of imported goods, indirect exposure through imported intermediate inputs used in domestic production, and domestic input–output linkages. Domestic wedges, however, create a disconnect between import prices at the border and producers’ costs, on the one hand, and retail prices, on the other, thereby dampening the response of CPI to exchange rate fluctuations. In particular, distribution costs—including transportation, insurance, and wholesaling—represent a substantial share of retail prices, reducing the effective weight of both imported and domestic producers’ costs in the CPI (Goldberg and Campa, 2010; Burstein et al., 2003). Variable markups further limit pass-through, as firms absorb part of the variation in marginal costs by adjusting markups rather than fully transmitting cost changes to prices (Klenow and Willis, 2016; Amiti et al., 2019). Finally, nominal rigidities in domestic pricing further attenuate retail price responses, as firms face frictions in adjusting prices to changes in marginal costs (Nakamura and Steinsson, 2008).

I apply the theoretical framework to Chile over the period 2008–2019, leveraging highly disaggregated data to discipline each margin of the model and assess the role of domestic wedges and their heterogeneity across products. To capture both direct and indirect exposure to imports, as well as the transmission of exchange rate changes through the domestic production network, I construct a detailed product-level (180×180) input–output table for the Chilean

¹ A key aspect of monetary policy trade-offs in open economies is determining which inflation measure is most relevant for policymakers, a decision shaped by the extent of exchange rate pass-through to domestic prices (Mishkin, 2008; Benigno and Benigno, 2003; Corsetti et al., 2010). Similarly, exchange rate fluctuations influence domestic redistribution as firms and consumers rely on varying mixes of domestic and imported goods (Cravino and Levchenko, 2017a; Jaravel, 2021). Understanding the relationship between the CPI and exchange rates, as well as the factors that shape it, also has important implications for the transmission of international shocks, the comovement of international business cycles, and the dynamics of external imbalances (Corsetti et al., 2008; Backus and Smith, 1993).

Figure 1: Estimated and Implied CPI Sensitivity



Note: The Figure compares the estimated CPI sensitivity to a one percent depreciation in the exchange rate with the implied sensitivity across a set of OECD countries. The blue circle represents the CPI sensitivity to exchange rate fluctuations estimated at the quarterly level over the period 1990–2019 (see Appendix B.4 for estimation details). The orange diamond depicts the implied CPI sensitivity, calculated as the sum of the share of imported final consumption goods and the share of imported inputs used in the production of domestic final consumption goods. The green line illustrates the range of implied CPI sensitivity when accounting for incomplete pass-through into import prices at the border. The upper and lower bounds reflect assumptions of 90% and 60% incomplete pass-through, respectively. Exchange rate and CPI data are sourced from the IMF, while the share of imports in the final consumption basket is computed using annual input-output tables from the OECD. Implied sensitivities are averaged over the 1990–2019 period.

economy. Each domestic wedge is calibrated allowing for product-level heterogeneity. Distribution costs are measured for each product as the difference between producer and basic prices, disaggregated by origin (domestic vs. imported) and use (intermediate vs. final consumption). Markups and their elasticities are estimated from firm-level data, assuming demand follows the Kimball specification in Klenow and Willis (2016). Nominal rigidities are calibrated at the product level using micro estimates of price adjustment frequencies from Pasten et al. (2024). Finally, the universe of import transactions is used to calibrate pass-through of exchange rate fluctuations into import prices, also accounting for heterogeneity across product categories.

I find that all domestic wedges are individually relevant and, collectively, more important than incomplete pass-through at the border in shaping the CPI response to exchange rate fluctuations. The benchmark model replicates the untargeted estimated sensitivity of the CPI, fully accounting for the gap between the blue and orange dots in Figure 1. This result highlights the importance of jointly modeling domestic wedges and incomplete pass-through at the border, while also validating the calibration strategy. Compared to a counterfactual economy in which exchange rate fluctuations are fully passed through to both import and domestic prices, domestic wedges reduce CPI sensitivity by 57 percent. In contrast, incomplete pass-through into import prices accounts for a 40 percent reduction. Each domestic wedge contributes meaningfully to this insensitivity: distribution costs alone reduce CPI sensitivity by roughly 36 percent, while variable markups and nominal rigidities account for reductions of 21 percent and

16 percent, respectively, as they operate only through the sensitivity of domestic final goods prices.

Product-level heterogeneity in wedges and composition effects in consumption dampen the aggregate response of CPI to exchange rate fluctuations. The model is calibrated at the product level, with sensitivity varying across products due to differences in domestic wedges, import exposure, and pass-through at the border. Abstracting from heterogeneity in wedges by setting them equal to the average across products raises CPI sensitivity by 55 percent relative to the benchmark. While joint heterogeneity across all wedges dampens CPI sensitivity, heterogeneity in individual wedges may either amplify or reduce it. In particular, removing heterogeneity in distribution costs and pass-through at the border lowers CPI sensitivity, whereas heterogeneity in nominal rigidities and variable markups amplifies it. In addition, heterogeneity in consumption shares further dampens CPI sensitivity, as expenditure is concentrated in goods less exposed to imported inputs. Finally, the aggregate effect of product-level heterogeneity can be rationalized by how wedges correlate with the underlying production structure—specifically, the share of imported inputs used in domestic production, which captures domestic goods’ exposure to exchange rate fluctuations.

Accounting for domestic wedges alters how exchange rate movements affect consumer prices. I show that wedges increase the importance of direct import exposure—imported goods consumed directly—while reducing the contribution of indirect exposure through imported intermediate inputs. The mechanism is that domestic wedges dampen the propagation of exchange rate shocks along the domestic input–output network, weakening spillovers through intermediates. As a result, direct exposure becomes the dominant driver of CPI sensitivity.

Beyond shifting the relative importance of transmission channels, wedges also reshape which products matter most for CPI sensitivity. The contribution of each sector to aggregate CPI pass-through depends on the interaction of heterogeneity in exposure, wedges, and consumption shares. Relative to a model without wedges or with homogeneous wedges, sectoral rankings change substantially, with some products moving by up to 80 positions. Which sectors emerge as most CPI-relevant depends critically on the set of wedges included, underscoring the importance of accounting for multidimensional heterogeneity when assessing exchange rate transmission.

Related Literature This paper contributes to the literature on the low sensitivity of domestic inflation to exchange rate fluctuations. Several studies have examined the role of domestic wedges. At the macroeconomic level, [Goldberg and Campa \(2010\)](#) quantify CPI sensitivity while accounting for import exposure and distribution costs across OECD economies. [Burstein et al. \(2003\)](#) and [Corsetti and Dedola \(2005\)](#) similarly show that distribution costs dampen the response of import and consumer prices to exchange rate changes. [Bacchetta and Van Wincoop \(2003\)](#) and [Atkeson and Burstein \(2008\)](#) focus on firms’ optimal pricing strategies and market structure. Building on these contributions, I integrate key, product-specific wedges into a unified theoretical framework and demonstrate their joint and individual quantitative relevance.

At the industry level, [Goldberg and Verboven \(2001\)](#) and [Hellerstein \(2004\)](#) decompose the sources of incomplete pass-through into non-traded costs and markup adjustments in the automotive and beer industries, respectively, while [Nakamura and Zerom \(2010\)](#) also account for nominal rigidities in the coffee industry. I extend this analysis from an industry-specific to an aggregate perspective, incorporating input-output linkages and sectoral heterogeneity, which significantly shape aggregate dynamics.

My work relates to the literature on production networks, heterogeneity in wedges, and the propagation of shocks.² [Rubbo \(2020\)](#) and [Pasten et al. \(2020\)](#) show, in a closed-economy setting, that heterogeneity in price rigidity is crucial for the transmission of monetary shocks. [Dhyne et al. \(2021\)](#) quantify the propagation of foreign demand shocks through domestic firm-to-firm transactions. Using Chilean data, [Huneus \(2018\)](#) examines the effects of foreign demand shocks in a model with an endogenous network. [Di Giovanni et al. \(2017\)](#), [Cravino and Levchenko \(2017b\)](#), and [Di Giovanni and Levchenko \(2010\)](#) investigate how multinational firms and international input-output linkages transmit productivity and inflation shocks across borders. Additionally, [Jones \(2011\)](#), [Bigio and La’o \(2020\)](#), and [Baqae and Farhi \(2019\)](#), among others, analyze the properties of inefficient networks with generic wedges. Relative to these studies, my analysis is complementary, focusing on the transmission of exchange rate fluctuations and the domestic wedges that shape their propagation.

An extensive body of work examines firm-level determinants of incomplete pass-through into border prices, including firm size and market share ([Berman et al., 2012](#); [Atkeson and Burstein, 2008](#)), reliance on imported inputs ([Amiti et al., 2014](#)), strategic complementarities ([Amiti et al., 2019](#)), product quality ([Chen and Juvenal, 2016](#)), bargaining and buyer market power ([Drozd and Nosal, 2012](#); [Heise, 2019](#); [Alviarez et al., 2021](#)), intra-firm versus arm’s-length relationships ([Neiman, 2010](#)), and invoicing choices ([Gopinath et al., 2010](#); [Chen et al., 2022](#)). I contribute to this literature by demonstrating that incomplete pass-through into border prices is quantitatively as significant as domestic wedges in shaping the overall sensitivity of CPI to exchange rate fluctuations, highlighting the aggregate relevance of firm-level determinants.

The rest of the paper is structured as follows. Section 1 presents the theoretical framework of pass-through, with a focus on key domestic wedges. Section 2 details the calibration strategy, and Section 3 reports the main results on the role of domestic wedges for the sensitivity of domestic prices to exchange rate fluctuations. Section 4 concludes.

1 A Model of Exchange Rate Pass-Through into CPI

This section outlines the theoretical framework used to analyze the role of domestic wedges in shaping the aggregate exchange rate pass-through to the consumer price index (CPI). The framework focuses on characterizing the domestic transmission of exchange rate fluctuations. I propose a parsimonious, static, multi-industry model, building on [Goldberg and Campa \(2010\)](#).

² See [Carvalho and Tahbaz-Salehi \(2019\)](#) for a recent survey.

The model incorporates several wedges that influence the domestic transmission of exchange rate changes and cost shocks in general, including distribution costs (Burststein et al., 2003; Corsetti and Dedola, 2005), variable markups (Goldberg and Verboven, 2001), and nominal rigidities (Gopinath and Itskhoki, 2011; Mukhin, 2022). Given the focus on domestic wedges, the response of import prices to exchange rate movements at the border is treated as exogenous. The framework highlights how heterogeneity in domestic wedges, along with differences in input-output linkages and the use of imported inputs in domestic production across sectors, shapes the domestic transmission of exchange rate fluctuations.

1.1 Environment

The section introduces the assumptions about preferences, production, and wedges. I then derive a set of equations to measure the exchange rate pass-through into CPI.

Price Aggregator. I consider a small open economy where local wages and prices are expressed in a national nominal unit of account. The economy is populated by a representative household whose preferences over consumption and labor are given by:

$$W(C, L) = U(C) - V(L), \quad (1)$$

where C and L represent the household's final consumption and total labor supply, respectively.³ The household consumes a set of N sectoral goods $i \in \{1, \dots, N\}$. Specifically, the household's final consumption basket, C , is represented by a homogeneous-of-degree-one consumption aggregator over sectoral goods, $C = \mathcal{C}(c_1, \dots, c_N)$. The household's utility maximization problem is subject to a standard budget constraint given by:

$$PC \equiv \sum_{i=1}^N p_i c_i \leq wL + \sum_{i=1}^{n_D} \int_0^1 \pi_{ik} dk, \quad (2)$$

where P is the nominal price index of the final consumption bundle, wL is labor income, and the last term captures the dividends from owning the domestic firms.

I assume that C takes the form of a Cobb-Douglas aggregator:

$$C(c_1, \dots, c_N) = \prod_{i=1}^N \left(\frac{c_i}{\beta_i} \right)^{\beta_i}, \quad \text{with } \sum_{i=1}^N \beta_i = 1 \quad (3)$$

where c_i is the amount of good i consumed, and the constants $\beta_i \geq 0$ represent the share of each good in the household's final consumption.

The utility-based final consumption price index, which serves as the model-implied measure

³ Typical regularity conditions are imposed on U and V : strictly increasing, twice differentiable, and $U'' < 0$, $V'' > 0$ and the Inada conditions are satisfied.

of the consumer price index, is given by:

$$P(p_1, \cdot, p_n) = \prod_{i=1}^N p_i^{\beta_i}, \quad (4)$$

where p_i is the retail price of the good of sector i , as paid by the household.

The pass-through of exchange rates into CPI, defined as the elasticity of CPI with respect to changes in the nominal exchange rate, e , and denoted by $\eta^{P,e}$, is expressed as:

$$\eta^{P,e} \equiv \frac{d \log P}{d \log e} = \boldsymbol{\beta} \times \boldsymbol{\eta}^{P,e}, \quad (5)$$

where $\boldsymbol{\beta}$ refers to the $N \times 1$ vector of expenditure shares, $(\beta_1, \dots, \beta_N)$, and $\boldsymbol{\eta}^{P,e}$ represents the $N \times 1$ vector of price elasticities, $(\eta^{p_1,e}, \dots, \eta^{p_N,e})^T$.

The pass-through of exchange rate movements into CPI is a weighted average of the pass-through rates into the prices of all goods consumed in the final consumption basket. Given the Cobb-Douglas specification in Equation (3), the relative weights correspond to the constant expenditure shares in total consumption, $\beta_i = \frac{p_i c_i}{PC}$.

The response of inflation to exchange rate fluctuations arises from both the direct and indirect consumption of imported goods, whose prices are directly influenced by exchange rate fluctuations. I assume that a subset n_D ($n_F = N - n_D$) of sectoral goods are produced domestically (imported from abroad). Thus, I disentangle the effects of direct and indirect import exposure, where the former refers to the presence of imported final consumption goods and the latter accounts for the use of imported intermediate inputs in the production of domestic goods. Formally, Equation (5) can be rewritten as:

$$\eta^{P,e} = \boldsymbol{\beta} \times \boldsymbol{\eta}^{P,e} = \underbrace{\boldsymbol{\beta}^D \times \boldsymbol{\eta}^{P^D,e}}_{\substack{\text{Indirect exposure} \\ \text{(Imported Intermediate Inputs)}}} + \underbrace{\boldsymbol{\beta}^F \times \boldsymbol{\eta}^{P^F,e}}_{\substack{\text{Direct exposure} \\ \text{(Imported Final Consumption)}}, \quad (6)$$

where $\boldsymbol{\eta}^{P^D,e}$ and $\boldsymbol{\eta}^{P^F,e}$ are the vectors of pass-through rates into the retail price of a domestically and imported sectoral goods, respectively.

In the following paragraphs, I first characterize the pass-through rate into the price of domestically produced goods, $\boldsymbol{\eta}^{P^D,e}$ in Equation (6), by introducing several elements that influence the transmission of exchange rate fluctuations. Then, I elaborate on the pass-through rate into the price of imported goods, $\boldsymbol{\eta}^{P^F,e}$.

Production of Domestic Goods - Local Distributor. I assume that each domestic sectoral good, $i \in n_D$, is produced by a local competitive distributor who aggregates a mass of sectoral varieties (La'O and Tahbaz-Salehi, 2022) and combines the aggregated good with local distribution services (Burstein et al., 2003). In turn, sectoral varieties are produced within each sector by a continuum of domestic monopolistically competitive firms, indexed by $k \in [0, 1]$.

The competitive distributor in each industry i aggregates the mass of differentiated varieties into an homogeneous sectoral good, y_i , using an homothetic Kimball aggregator, [Kimball \(1995\)](#):

$$\sum_k A_{ik} \mathcal{K}_i \left(\frac{y_{ik}}{y_i} \right) = 1, \quad (7)$$

where y_{ik} is the consumption of variety k in industry i , and A_{ik} is a quality parameter; $\mathcal{K}(\cdot)$ is such that $\mathcal{K}(\cdot) > 0$, $\mathcal{K}'(\cdot) > 0$, $\mathcal{K}''(\cdot) < 0$ and $\mathcal{K}(1) = 1$. The distributor's VES technology represents the demand schedule faced by monopolistically competitive firms. In the quantitative analysis in Section 2, I adopt the common [Klenow and Willis \(2016\)](#) formulation for the Kimball aggregator. In this case, Marshall's weak second law holds, implying that as firms lower their prices, their demand becomes more inelastic and their markup increases. Thus, larger firms will exhibit higher markups, higher markup elasticity, and a lower pass through rate of cost shocks ([Atkeson and Burstein, 2008](#); [Kimball, 1995](#)).⁴

The distributor sells the homogeneous sectoral good y_i , incurring in distribution costs. These costs represent the per-unit service inputs required to deliver the sectoral goods to consumers and firms, e.g. transportation, wholesales and retail services, marketing, etc ([Burstein et al., 2003](#); [Corsetti and Dedola, 2005](#)). I assume that distribution services are combined with one unit of sectoral good using a Cobb-Douglas technology, and that distribution services are produced using labor ([Atkeson and Burstein, 2008](#)). Thus, the retail price of good i , p_i , is:

$$p_i = \tilde{p}_i^{\phi_i} w^{1-\phi_i} \quad \text{with } \phi \leq 1, \quad (8)$$

where $\tilde{p}_i$ is the price of the sectoral good i , and $1 - \phi_i$ represents the share of distribution services in the retail price of good i . I assume that distribution costs are heterogeneous across sectors, as indicated by the i -specific weights in the production technology.

Production of Domestic Goods - Sectoral Varieties. The monopolistically competitive firms within each domestic industry use a firm-specific, constant returns to scale production function. Both domestic and imported sectoral goods can be used as inputs in the production of domestic varieties, along with labor. Specifically, I assume that firms employ the following Cobb-Douglas technology:

$$y_{ik} = F_i(l_{i,k}, x_{i1,k}, \dots, x_{iN,k}) = \zeta_{ik} l_{i,k}^{\alpha_{ik,l}} \prod_{j=1}^N x_{ij,k}^{\alpha_{ik,j}} \quad \text{with } \alpha_{ik,l} + \sum_{j=1}^N \alpha_{ik,j} = 1 \quad \forall k, \quad (9)$$

where $y_{i,k}$ is firm k 's output, ζ_{ik} is the firm's productivity parameter, $l_{i,k}$ is the labor input, $x_{ij,k}$ is the amount of good j used as input by firm k in sector i , and $\alpha_{ik,j} \geq 0$ denotes the share

⁴ [Amiti et al. \(2019\)](#), among others, shows that monopolistically competitive firms with Kimball demand generate qualitatively similar predictions regarding pass-through and heterogeneity to a model of oligopolistic firms with nested CES demand, as in [Atkeson and Burstein \(2008\)](#). I favor the former specification because its calibration strategy better aligns with the available data.

of good j in the production of firm k in sector i .⁵

Each firm potentially has heterogeneous direct and indirect exposure to exchange rate fluctuations due to its use of a different mix of inputs, as reflected by the firm-specific exponents, $\alpha_{ik,j}$. A firm's production cost is directly exposed to exchange rate fluctuations when it directly uses imported inputs in production. However, a firm may also be exposed indirectly, even when it does not use any imported input. This occurs through domestic input-output linkages to other domestic firms that employ imported inputs.

Given the assumption about the distributor's aggregating technology, monopolistically competitive producers charge a variable markup over marginal cost:

$$\widetilde{p}_{ik} = \mu_{ik} mc_{ik} \quad \text{with } mc_{ik} = \frac{1}{\zeta_{ik}} w^{\alpha_{ik,l}} \prod_{j=1}^N p_j^{\alpha_{ik,j}}, \quad (10)$$

where $\widetilde{p}_{ik}$ is the price paid by the distributor for variety k , μ_{ik} is the markup charged, and the expression for the marginal cost, mc_{ik} , follows from the specific production function assumed in Equation (9). Given the functional form in Equation (7), the markup charged by monopolistically competitive firms increases in firm sales and becomes more sensitive to cost shocks, implying a lower pass through rate.

Lastly, I assume that monopolistically competitive producers are subject to nominal rigidities. Producers set the price of their variety based on their marginal cost before the realization of shocks. I use a one-period version of Calvo (1983) pricing, assuming that producers can update their price with a sector-specific probability δ_i (Mukhin, 2022; Rubbo, 2020). Thus, a fraction δ_i of firms in sector i can adjust their prices to changes in their marginal costs following exchange rate fluctuations.

Imported Goods. I assume that imported sectoral goods ($i \in n_F$) are produced abroad and purchased by a local distributor, who combines them with domestic distribution services prior to sale. Accordingly, as shown in Equation (8), the domestic retail price of imported goods, p_i , is given by:

$$p_i = (\widetilde{p}_i(e))^{\phi_i} w^{1-\phi_i}, \quad \text{with } \phi_i \leq 1, \quad (11)$$

where $i \in n_F$, and $\widetilde{p}_i(e)$ denotes the local-currency producer price of the imported good. Under the small open economy assumption, the producer price is determined by the foreign supplier and taken as given by the local distributor. The local-currency price $\widetilde{p}_i$ is influenced by the nominal exchange rate. The exchange rate pass-through to import prices is not microfounded in the model, but rather treated as exogenous and estimated empirically from the data.

⁵ I assume that $\alpha_{ik,l} > 0$, i.e. that labor is an essential input for the production of all varieties, in the sense that $F_{ik}(0, x_{i1,k}, \dots, x_{iN,k}) = 0$.

1.2 Theoretical Results

This section develops intuition for the response of domestic prices to exchange rate fluctuations, considering sectoral heterogeneity in wedges and exposure, within and across sector heterogeneity, and input-output linkages. Further details are reported in the Appendix A.1.

Marginal Cost and Sectoral Prices. The behavior of marginal costs is crucial for understanding the response of domestic prices to exchange rate fluctuations. I focus on the direct effect of exchange rate, excluding changes in the wage rate or the response of firms to changes in sectoral price indices and quantities (Burstein and Gopinath, 2014).

At the firm level, the change in the price of variety k in a domestic sector i can be written:

$$\partial \log \widetilde{p}_{ik} = \delta_i \kappa_{ik} \partial \log mc_{ik}, \quad (12)$$

where $\kappa_{ik} \equiv \frac{1}{1+\Gamma_{ik}} < 1$, with $\Gamma_{ik} \equiv -\frac{\partial \log \mu_{ik}}{\partial \log \widetilde{p}_{ik}} > 0$ being the markup elasticity, and δ_i the sector-specific Calvo probability. Equation (12) illustrates how two wedges - variable markup and nominal rigidities - reduce the response of prices to changes in marginal costs. While nominal rigidities hinder the adjustment of prices, variable markups make firms less inclined to adjust their prices relative to other firms' prices, leading them to absorb a portion of the cost changes instead.⁶

Given the production technology, the change in the marginal cost of producer k in domestic sector i following a change in input prices is as follows:

$$\partial \log mc_{ik} = \sum_{j=1}^N \alpha_{ik,j} \partial \log p_j = \sum_{j=1}^{n_D} \alpha_{ik,j} \partial \log p_j + \sum_{j=1}^{n_F} \alpha_{ik,j} \partial \log p_j(e), \quad (13)$$

where $\alpha_{ik,j}$ is the exposure of firm k in sector i to input j , and p_j represents the retail price of input j . The second part of the equation highlights how a firm's marginal cost can be influenced directly by exchange rate fluctuations through the use of imported inputs, as well as indirectly through the use of domestic inputs. This emphasizes the role of input-output linkages in shaping marginal costs, and consequently, in affecting the transmission of exchange rate fluctuations to domestic prices.

We are interested in characterizing the response of sectoral prices to exchange rate fluctuations, aggregating up the price responses of individual firms in Equation (12):

$$\partial \log \widetilde{p}_i \equiv \sum_{k \in N_i} S_{ik} \partial \log \widetilde{p}_{ik} = \sum_{k \in N_i} S_{ik} \delta_i \kappa_{ik} \partial \log mc_{ik} = \delta_i [\bar{\kappa}_i \partial \log mc_i + cov_{S_{ik}}(\kappa_{ik}, \partial \log mc_{ik})], \quad (14)$$

where S_{ik} is the revenues share of firm k in sector i ; $\bar{x}$ is the revenues-weighted average of variable x , i.e. $\bar{x} = \sum_{k \in N_i} S_{ik} x_{ik}$; and $cov_{z_{ik}}(x_{ik}, y_{ik})$ is the z -weighted covariance between two variables x and y , i.e. $cov_{z_{ik}}(x_{ik}, y_{ik}) = \sum_{k \in N_i} z_{ik} (x_{ik} - \bar{x})(y_{ik} - \bar{y})$.

⁶ κ_{ik} formally measures the incomplete degree of pass-through of cost shocks into prices. The pass-through rate inversely depends on how much the markup is sensitive, i.e. on the markup elasticity Γ_{ik} .

The last part of Equation (14) shows that the aggregate sectoral response to exchange rate fluctuations can be decomposed into two components: an average effect and a heterogeneous effect. The average effect is captured by the revenue-weighted average response across firms, measured by the average change in the marginal cost muted by the average incomplete pass-through rate due to variable markups ($\bar{\kappa}_i$). The heterogeneous effect, on the other hand, depends on the within-sector revenue-weighted correlation between markup elasticity and marginal cost fluctuations, highlighting the potential relevance of within-sector systematic heterogeneity in markup elasticity and exposure (Amiti et al., 2019).

Moreover, by leveraging the definition of marginal cost and the economy's production structure, I can further specialize Equation (14) in terms of input exposure:

$$\partial \overline{\log mc_i} = \sum_{j=1}^N \bar{\alpha}_{ij} \partial \log p_j, \quad cov_{S_{ik}}(\kappa_{ik}, \partial \log mc_{ik}) = \sum_j^N \partial \log p_j cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j}), \quad (15)$$

where $\bar{\alpha}_{ij} = \sum_{k \in N_i} S_{ik} \alpha_{ik,j}$ is the revenue-weighted average input share of product j in sector i . Thus, the response of sectoral prices depends on the average exposure of the sector to imported and domestic inputs ($\bar{\alpha}_{ij}$), muted by the presence of variable markups and nominal rigidities.

Within-sector heterogeneity can either dampen or amplify the response of sectoral prices, depending on the correlation between exposure and size at the firm level. Under the assumptions of the distributor's Kimball aggregator in Equation (7), markup elasticity increases with firm size, while pass-through decreases. If larger firms are also more exposed—either directly or indirectly—to exchange rate fluctuations, the covariance is negative. Consequently, the response of sectoral prices is smaller than the average effects, as larger firms exhibit lower pass-through. However, I do not impose any ex-ante correlation between exposure and firm size, leaving this relationship to be determined during the calibration in Section 2.

Aggregate Exchange Rate Pass-through in Retail Prices. I present closed-form expressions for the response of inflation to exchange rate fluctuations. Under the assumptions regarding the distribution sector in Equation (8), the change in the retail price of good i can be expressed as follows:

$$\partial \log p_i = \phi_i \partial \log \tilde{p}_i, \quad (16)$$

where the presence of distribution costs further dampens the response of retail prices to changes in production costs (e.g. due exchange rate fluctuations), as production costs account for only a fraction ϕ_i of the retail price.

Combining these equations across all sectors with Equations (14) and (15), the $n_D \times 1$ vector

of pass-through rates into the retail prices of domestic goods, $\boldsymbol{\eta}^D = \left[\frac{\partial \log p_1}{\partial \log e}, \dots, \frac{\partial \log p_H}{\partial \log e} \right]^T$ is:

$$\boldsymbol{\eta}^D = \underbrace{\left[\mathbf{I} - \underbrace{\boldsymbol{\Phi}^D \boldsymbol{\Delta}^D (\boldsymbol{\Gamma}^D \mathbf{S}^D + \boldsymbol{\Lambda}^D)}_{\text{Adjusted Leontief}} \right]^{-1}}_{\text{IO network}} \underbrace{\boldsymbol{\Phi}^D \boldsymbol{\Delta}^D \left(\underbrace{\boldsymbol{\Gamma}^D \mathbf{S}^F}_{\text{Average}} + \underbrace{\boldsymbol{\Lambda}^F}_{\text{Heterogeneous}} \right)}_{\text{Import Exposure}} \underbrace{\boldsymbol{\Phi}^F \boldsymbol{\Psi}^F}_{\boldsymbol{\eta}^F}, \quad (17)$$

where $\boldsymbol{\eta}^F$ and $\boldsymbol{\Psi}^F$ are $n_F \times 1$ vectors of pass-through rate into the retail price and the producer price of imported goods, respectively; $\boldsymbol{\Phi}^F$ and $\boldsymbol{\Phi}^D$ are the $n_F \times n_F$ and $n_D \times n_D$ diagonal matrices of sectoral distribution margins; $\boldsymbol{\Delta}^D$ is the $n_D \times n_D$ diagonal matrix of sectoral Calvo parameters; $\boldsymbol{\Gamma}^D$ is a $n_D \times n_D$ matrix of sectoral average markup elasticities $\bar{\kappa}i$; $\boldsymbol{\Lambda}^F$ and $\boldsymbol{\Lambda}^D$ are the $n_D \times n_F$ and $n_D \times n_D$ matrices collecting the within-sector correlations between markup elasticities and the use of imported and domestic inputs in the production of domestic goods, respectively; $\mathbf{S}^D$ is the $n_D \times n_D$ domestic Leontief matrix, where each element $\bar{\alpha}_{ij}$ represents the share of domestic product j used as an input by domestic sector i ; similarly $\mathbf{S}^F$ is the $n_D \times n_F$ international Leontief matrix collecting the expenditure share of domestic sector i on each imported goods j . The vectors of pass-through rates into the retail price of domestic and imported goods ($\boldsymbol{\eta}^D$ and $\boldsymbol{\eta}^F$, respectively) fully characterize the elements determining the transmission of exchange rate fluctuations into the CPI, as outlined in Equation (6). Additional details on the derivations in Appendix A.1.

The final term in Equation (17) illustrates that the pass-through of exchange rate fluctuations into the retail price of domestically produced goods, $\boldsymbol{\eta}^D$, depends on the pass-through of exchange rate fluctuations into the local price of imported inputs, $\boldsymbol{\eta}^F$. This, in turn, is determined by the sensitivity of the local-currency producer price of the imported good to exchange rate fluctuations, $\boldsymbol{\Psi}^F$, weighted by the share of local distribution services, $\boldsymbol{\Phi}^F$, as specified in Equation (11).

$\boldsymbol{\eta}^D$ depends not only on changes in the price of imported inputs but also on domestic firms' exposure to these inputs and the presence of domestic wedges. As shown in Equation (15), the second-to-last term consists of two components: an average effect - representing the average use of imported inputs, moderated by the variable markup wedge - and a heterogeneity effect, which captures the correlation between the use of imported inputs and markup elasticity within a sector. Similarly, the impact of the other two wedges, nominal rigidities and distribution services, in dampening the overall pass-through is captured by the multiplicative term $\boldsymbol{\Phi}^D \boldsymbol{\Delta}^D$.

The presence of input-output linkages amplifies the pass-through of exchange rate fluctuations on domestic prices, although domestic wedges dampen these amplification effects. Input-output linkages are captured by the first term, the adjusted Leontief inverse matrix, which depends on the input-output matrix of domestic inputs $\mathbf{S}^D$. However, this form of indirect exposure is adjusted to account for the presence of domestic wedges, as the actual response of domestic prices is dampened by nominal rigidities, variable markups, and distribution services.

On Heterogeneity. The contribution of each sector and good to the aggregate transmission of exchange rate fluctuations into the CPI depends on the interplay of heterogeneities in exposure, wedges, and consumption shares across sectors. For example, Equation (6) shows that the correlation between consumption shares and pass-through rates at the sector level can either amplify or dampen the aggregate transmission. If final consumers allocate larger shares of their consumption basket onto goods that are more sensitive to exchange rate fluctuations (i.e. goods with higher pass-through rates), the aggregate pass-through rate to CPI is higher.

The contribution of a sector to the aggregate transmission of exchange rate fluctuations into CPI also depends on the interaction between the intensity of imported input usage and the strength of domestic wedges. Stronger domestic wedges — such as higher distribution costs, more rigid nominal prices, or greater markup elasticities—exert a larger dampening effect on the aggregate transmission when they are positively correlated with sectors that heavily rely on imported inputs. These sectors are typically the most significant channels through which exchange rate fluctuations affect the CPI.

Similarly, a sector’s contribution depends on its centrality in the domestic input-output network, adjusted for the presence of domestic wedges. Sectors producing goods that are extensively used as inputs in the production of other goods contribute more to the aggregate transmission of exchange rate fluctuations into CPI. However, the aggregate dampening effects of domestic wedges are amplified when these central sectors face stronger wedges.

The presence and interaction of these various sources of heterogeneity represent an example of the identity effect (Pasten et al., 2020). The degree to which a sector transmits exchange rate fluctuations to the CPI depends on the precise interaction of these heterogeneities. I impose no restrictions on the joint distribution of these factors, leaving the quantitative effect to the empirical section below. The ranking of importance across sectors may shift significantly when one or more sources of heterogeneity are abstracted away, underscoring the need for a comprehensive analysis. From a policy perspective, the identity effect is key, as optimal targeting or redistribution policies require precise knowledge of the most relevant sectors.

1.3 Discussion

I conclude this section by highlighting and discussing the key theoretical assumptions.

The assumption of perfect competition in the distribution sector can be relaxed without altering the key theoretical insights, provided that distributors charge constant markups. Allowing for variable markups would permit distributors to adjust their margins in response to exchange rate fluctuations, introducing an additional source of incomplete pass-through. Empirically, however, Goldberg and Campa (2010) show that distribution margins decrease only weakly following an exchange rate depreciation. From a theoretical point of view, models with additive distribution costs (Corsetti and Dedola, 2005) or Leontief technology (Ruane and Peter, 2022) yield the same qualitative implications for pass-through as the Cobb-Douglas technology. Calibration is more straightforward under the approach of Atkeson and Burstein (2008), as the shares ϕ_i can be computed directly from the input–output tables.

One key assumption is separability between domestic and foreign suppliers, which allows the analysis to abstract from any micro-foundations of foreign production and pricing of imported goods, treating their response to exchange rate fluctuations as exogenous. While interactions between domestic and foreign suppliers matter for markup and concentration dynamics (Amiti and Heise, 2024), the data used in the calibration do not allow me to assess such dynamics. Nevertheless, the granularity of transaction-level customs data used in the quantitative analysis permits direct measurement of the response of import prices to exchange rate fluctuations, thereby allowing some of these general equilibrium effects to be absorbed.⁷

Lastly, an important assumption concerns the functional form of the production function, which imposes unit elasticity of substitution. This assumption may imply relatively high expenditure switching and substitutability across goods, particularly given the focus on short-term pass-through.⁸ A unit elasticity lies within the range of estimates in international economics using high-frequency variation from transitory shocks (Ruhl et al., 2008), but it is at the upper end of the estimates in the macroeconomic literature, which range from 0.3 in Atalay (2017) and Oberfield and Raval (2021) to 1.5 in Adão et al. (2022) and Carvalho et al. (2021), and slightly above the values typically found in the trade literature (around 0.8 (Boehm et al., 2023)).⁹ From a theoretical perspective, a unit-elasticity framework provides, on the one hand, a first-order approximation to richer substitution patterns (Hulten, 1978; Baqaee and Farhi, 2020; Carvalho and Grassi, 2019), and on the other hand, a particularly simple and insightful expressions for the role of wedges and distortions in networked production structures (Grassi et al., 2017; La’O and Tahbaz-Salehi, 2022).

2 Calibration

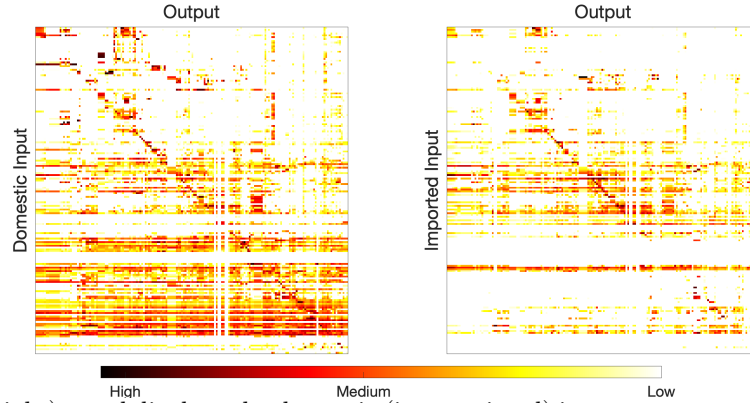
Each element in Equations (6) and (17), along with their heterogeneity across products, is calibrated using a variety of micro-level datasets. The primary inputs include the granular ”make” and ”use” tables from the Central Bank of Chile (2008–2019), data from the Annual Survey of Manufacturing (ENIA, Encuesta Nacional Industrial Anual) compiled by the Chilean National Statistical Agency (INE, Instituto Nacional de Estadísticas) for the period 1995–2015, and the universe of Chilean import transactions (2007–2019) from the Chilean Customs Agency (Aduanas). Below, I detail the strategy employed to calibrate each component of the main measurement equations. Additional information is provided in Appendix B.

⁷ Separability is consistent with the small open economy assumption, which rules out international input–output linkages: changes in the price of domestic goods do not affect the foreign production costs of imported goods, thereby excluding roundabout effects.

⁸ The framework could potentially speak to the issue of long-run exchange rate pass-through, however it would require taking an explicit stance on the conduct and objectives of monetary policy—such as whether the central bank targets inflation strictly, operates under a Taylor rule or engages in discretionary stabilization, whether it explicitly takes into account exchange rate stability.

⁹ In the international real business cycle literature, matching observed terms-of-trade volatility and the negative relationship between terms of trade and trade balance typically requires low trade elasticity values (Hillberry and Hummels, 2013).

Figure 2: Domestic and International Leontief Matrices



Note: The left (right) panel displays the domestic (international) input-output matrix of the Chilean economy, with shares averaged over the period 2013-2019. The matrices are computed using the "make" and "use" tables under the industry technology assumption. Each row (column) represents an input (output). The intensity of the coloring indicates the extent to which one product is used as an input in the production of other products. Darker colors correspond to higher input shares, while lighter colors indicate lower input shares. The log of input shares below -10 are censored.

Leontief Matrices and Consumption Shares I construct the input-output matrices for the Chilean economy from 2008 to 2019 by combining the "make" and "use" tables provided by the Central Bank of Chile. These tables consist of two basic national accounting components: the "make" table, which shows the production of goods by sector, and the "use" table, which shows the use of goods by sectors and final users. The Central Bank of Chile also provides information on international flows, allowing the construction of international "make" (for imports) and "use" (for exports) tables. The tables are highly disaggregated and include 110 industries and 175 products before 2013, and 180 products starting on 2013.¹⁰

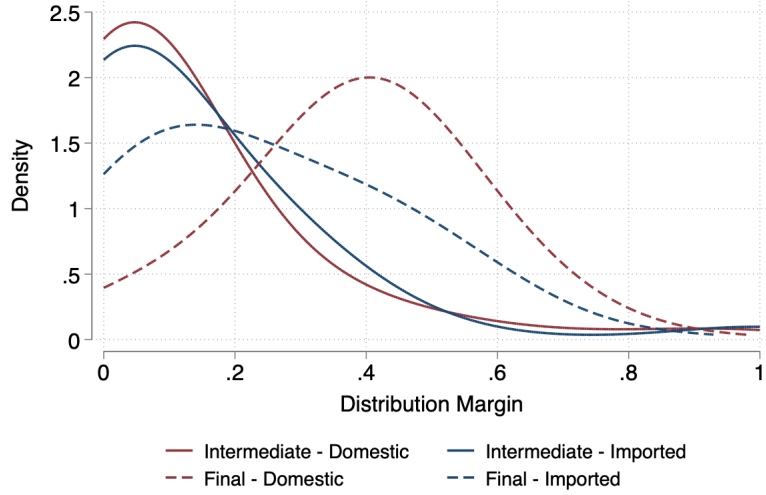
I combine the "make" and "use" tables under the industry technology assumption to construct a product-by-product input-output matrix of dimension 175×175 or 180×180 .¹¹ Each matrix quantifies the extent to which each product (row) is used in the production of other products (column). This structure allows me to calibrate each element $\bar{\alpha}_{ij}$ of the domestic and international Leontief matrices, $\mathbf{S}^D$ and $\mathbf{S}^F$. Figure 2 displays the average Leontief matrices over the period 2013-2019, where darker colors indicate a higher share of a given input in the production of a specific product. Domestic network and trade exposure are highly sparse and heterogeneous across products, characteristics that play a crucial role in shaping the response of aggregate variables (Pasten et al., 2020).¹² Lastly, I use the "use" tables to compute the share of each product in final consumption, calibrating the vector β .

¹⁰ For comparison, commonly used input-output tables, such as the WIOD or the OECD tables, typically contain around 30 to 40 sectors. Pasten et al. (2020) shows that the granularity of the input-output table plays a central role in quantifying the real effects of monetary policy, as less granular input-output tables tend to underestimate these effects.

¹¹ Appendix B provides details on the technical assumptions used in the construction of the input-output matrices.

¹² Figure 15 in Appendix B shows the average Leontief matrices over the period 2008-2012, with qualitatively similar patterns.

Figure 3: Distribution Margins



Note: The Figure plots the density distribution of the distribution margins across products with positive use shares. The distribution margins are computed using Equation (18) and input-output tables from 2008 to 2019. Products are differentiated based on their use (final vs. intermediate use), with solid lines representing final goods and dashed lines representing intermediate goods. Additionally, products are classified by their origin, with blue lines for imported goods and red lines for domestically produced goods.

Distribution Margin Following Goldberg and Campa (2010), I compute the product-level distribution margin $1 - \phi_i$ as the ratio of the value of trade and transport margins to the value of total supply of that product at purchasers' prices:

$$1 - \phi_i \equiv \frac{\text{Retail} + \text{wholesale} + \text{transportation costs}}{\text{Value at purchaser prices}} \equiv \frac{\text{Value at purchaser prices} - \text{value at basic prices}}{\text{Value at purchaser prices}}. \quad (18)$$

I use the input-output matrices for the Chilean economy to compute the value of trade and transport margins as the difference between the cost of supply (basic price) and the purchaser price. The data allows me to calculate time-varying, heterogeneous margins across products, as well as variations across use (final vs. intermediate consumption) and origin (imported vs. domestic).

Figure 3 reports the density distribution of distribution margins across different product categories (domestic vs. imported, intermediate vs. final). The distribution margin is found to be large and highly heterogeneous across products, origins, and uses. On average, domestically produced goods have lower distribution margins compared to imported goods (18% vs 26%, respectively), which is consistent with the higher transportation costs associated with internationally sourced products. Moreover, intermediate goods generally have lower distribution margins than final goods (13% vs 31%, respectively).¹³ This suggests that lower pass-through due to distribution costs may arise primarily at the end of the production chain, when goods reach final consumers. Panel B of Table 4 in Appendix B highlights considerable heterogeneity across sectors: distribution margins in manufacturing sectors such as food and textiles are approximately 25%-30%, while in non-tradable sectors such as communication, health care and education are around 8% (Burstein et al., 2003; Berger et al., 2012).

¹³ Values are reported in Panel A of Table 4 in Appendix B.

Markup Elasticity Operationalizing the theoretical framework in Equation (17) requires estimating firm-level markup elasticities to calibrate the matrices $\mathbf{\Gamma}^D$, $\mathbf{\Lambda}^D$, and $\mathbf{\Lambda}^F$. In order to do so, additional assumptions on the functional form of the Kimball aggregator in Equation (7) are required. These assumptions enable the parameterization of markup elasticities Γ_{ik} and, ultimately, the firm-level pass-through rates κ_{ik} used to construct $\mathbf{\Gamma}$ and $\mathbf{\Lambda}$ s. I assume that the Kimball aggregator in Equation (7) takes the form of a sector-specific [Klenow and Willis \(2016\)](#) aggregator and calibrate it following [Edmond et al. \(2018\)](#). Under the [Klenow and Willis \(2016\)](#) specification, the markup elasticity of firm k in sector i , $\Gamma_{ki} = -\frac{\partial \log \mu_{ki}}{\partial \log p_{ki}}$, takes the following function form:

$$\Gamma_{ki} = \frac{\epsilon_i}{\sigma_i} \mu_{ki}, \quad (19)$$

where $\sigma_i > 1$ and ϵ_i represent the sector-specific demand elasticity and the demand superelasticity, respectively. Thus, the markup elasticity can be computed using σ_i , ϵ_i , and firm-level markups μ_{ki} .

Firm-level pass-through rates are estimated in two steps. First, I use firm-level balance sheet data from the Annual Survey of Manufacturing (ENIA), administered by the Chilean National Institute of Statistics (INE), spanning the years 1995 to 2015, to compute firm-level markups following [Autor et al. \(2020\)](#). Under the assumption of constant returns to scale, markups are measured as the ratio of firm sales to total costs:

$$\mu_{kit} = \frac{\alpha_{kit}^v}{S_{kit}^v} = \frac{\text{Sales}_{kit}}{\text{Total Cost}_{kit}}, \quad (20)$$

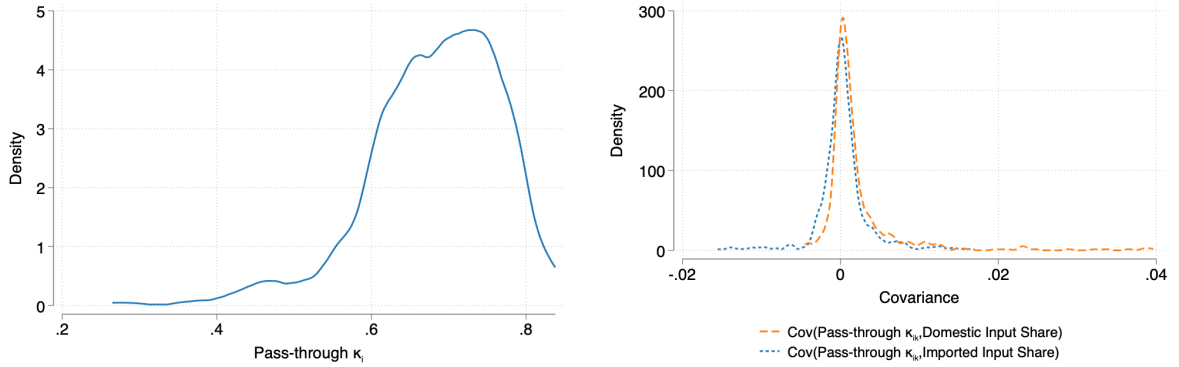
where α_{kit}^v and S_{kit}^v denote the output elasticity and the factor share of input v , respectively ($S_{kit}^v = \frac{\text{Expenditure on } v}{\text{Sales}_{kit}}$). The second equality arises from the CRS assumption, which implies $\alpha_{kit}^v = \frac{\text{Expenditure on } v}{\text{Total Cost}_{kit}}$. I assume that total costs are the sum of wage bill, materials expenditure, and capital costs. In the second step, I jointly estimate the elasticity and super-elasticity of demand following [Edmond et al. \(2018\)](#). The [Klenow and Willis \(2016\)](#) specification allows to specify the following within-industry relationship between markups and market shares:

$$\frac{1}{\mu_{kit}} + \log \left(1 - \frac{1}{\mu_{kit}} \right) = \text{constant} + \gamma \log s_{kit} + \nu_{kit} \quad (21)$$

where $\gamma = \frac{\epsilon_i}{\sigma_i}$ and s_{kit} is the market share of firm k in sector i . I estimate this specification for each sector in the economy (CIIU industry), introducing firm and year fixed effects to control for unobserved productivity, quality differences, and aggregate dynamics.

The sectoral estimates are then mapped to the input-output product categories to construct the matrices $\mathbf{\Gamma}^D$, $\mathbf{\Lambda}^D$, and $\mathbf{\Lambda}^F$ in Equation (17), after creating a concordance between CIIU classification and input-output product categories. After estimating firm-level pass-through rates κ_{ik} , I can compute the revenue-weighted average pass-through rate ($\bar{\kappa}_i$) for each sector i and construct the matrix $\mathbf{\Gamma}^D$. The covariance matrices $\mathbf{\Lambda}^D$ and $\mathbf{\Lambda}^F$ require information on how much of each specific input j is used by firms. However, the ENIA balance sheet data only distinguishes between total domestic and total imported in-

Figure 4: Pass-Through Rate and Covariance Term



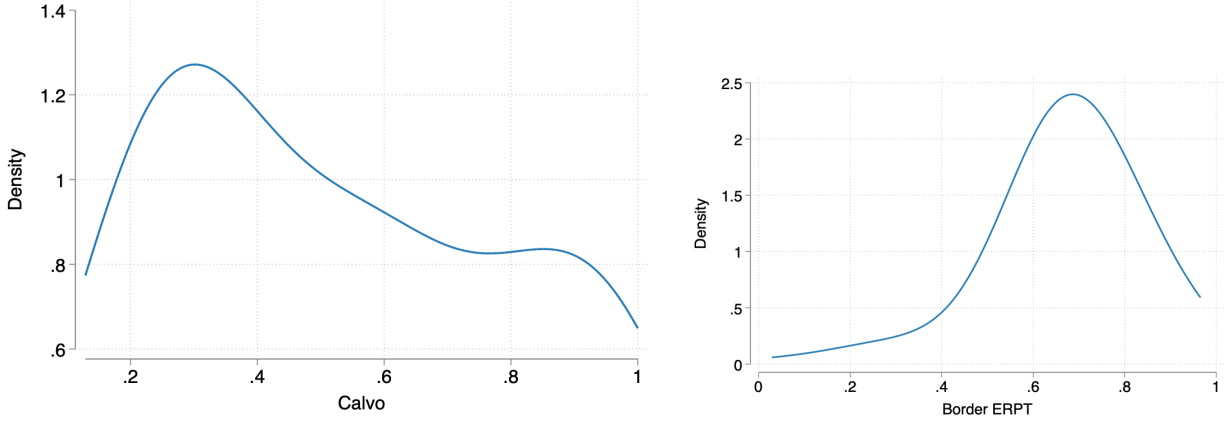
Note: The blue solid line on the left panel plots the distribution of revenue-weighted average pass-through rate $\bar{\kappa}_i$ across sectors. The green and orange dash lines on the right panel represent the covariance between pass-through rates and the aggregate share of domestic and imported inputs, respectively. Firm-level pass-through rates and input shares are estimated using firm-level balance sheet data from the Annual Survey of Manufacturing (ENIA), administered by the Chilean National Institute of Statistics (INE), spanning the years 1995 to 2015. Appendix B provides additional details on data and estimation.

put use at the firm level—it does not provide input-level detail within these two categories. To address this data limitation, I assume that within a given sector i , the covariance between a firm’s pass-through rate κ_{ik} and its share of any individual input j is the same across all inputs. Formally, this means: $cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j}) \equiv cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j'}) \quad \forall j, j'$. This simplifying assumption allows me to use the linearity of covariance to recover the common covariance term in terms of aggregate share of domestic and imported inputs. Specifically, the covariance between pass-through and the aggregate share of domestic inputs equals the sum of the covariances between pass-through and the shares of individual domestic inputs: $cov_{S_{ik}}(\kappa_{ik}, \text{Share of domestic inputs}_{ik}) = \sum_{j \in D} cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j})$. Since all covariances are assumed equal across j , the common covariance term can be computed by dividing the aggregate covariance by the number of domestic inputs used in sector i , which is available from the input-output tables. A similar reasoning applies to the covariance for imported inputs. Appendix B provides additional details on data construction, estimation, and robustness exercises.

Figure 4 displays the distribution of the revenue-weighted average pass-through rate ($\bar{\kappa}_i$) and the covariance between pass-through rates and the shares of domestic and imported inputs across sectors. There is substantial heterogeneity in the average pass-through rate, which ranges from 0.3 to 0.85. The covariance terms are small in magnitude, ranging from -0.01 to 0.01 , and include both positive and negative values, indicating qualitative differences in the relationship between pass-through and the intensity of domestic and imported input use across sectors. However, their quantitative significance depends on their distribution and how they interact with other distortions in the economy.¹⁴

¹⁴ Table 3 in Appendix B reports key moments of the estimated markup elasticities and pass-through rates at the firm-level. The estimated values are in line with previous work (Amiti et al., 2019; Baqaee and Farhi, 2020). On average, firms with higher pass-through rates exhibit a larger share of domestic inputs (Figure 12 in Appendix B).

Figure 5: Calvo Adjustment and Border ERPT



Note: The left panel plots the distribution of the Calvo frequencies of price adjustment at quarterly level across product categories in the Chilean input-output tables. These are constructed from the PPI BLS data (Pasten et al., 2024). The right panel plots the distribution of exchange rate pass-through rates on border prices at quarterly frequency across product categories in the Chilean input-output tables. These are estimated using Chilean customs records from 2007 to 2019 and the specification in Equation (22) at the HS2 level.

Nominal Rigidities I calibrate the product specific Calvo frequency of price adjustment using information available from Pasten et al. (2024). They use the confidential microdata underlying the PPI of the BLS to calculate the frequency of price adjustment at the industry level, defined as the ratio of the number of price changes to the number of sample months. I construct a concordance table between NAICS and the Chilean input-output product classification and then rescale the original monthly frequency of price adjustment to the quarterly level. The average duration is of 2.04 quarters (6.13 months), implying an average frequency of price adjustment of 0.53. Substantial heterogeneity is present in the frequency across products, with raw materials and farming product exhibiting durations lower than 0.3 quarters and many service industries showing durations above 4.5 quarters (left panel of Figure 5).¹⁵

Import Price Sensitivities I use Chilean Custom records from 2007 to 2019 to calibrate product-level pass-through rates of exchange rate fluctuations into import prices at the border. The customs data records, for each import transaction, information such as the importer's unique identifier, the 8-digit HS product code, the transaction date, the country of origin, the value, and the quantity shipped. Appendix B provides additional details on cleaning and summary statistics.

I estimate a standard reduced-form exchange rate pass-through regression to measure the transmission of exchange rate fluctuations into import prices at the border (Burstein and Gopinath, 2014). The pass-through is estimated at quarterly frequency to be consistent with the Calvo frequency of adjustment Δ^D . Let f index an importing firm, p an HS8 product

¹⁵ As a robustness, I rescale the distribution of the frequency of price adjustments to match the average monthly frequency of price adjustment of 30%, as estimated by Aruoba et al. (2022) from confidential daily transaction data from the Chilean Tax Authority. This implies an average quarterly probability of adjustment of 0.65, slightly reducing the effect of nominal rigidities on the exchange rate pass-through into CPI.

category, o the country of origin, and t the quarter. The baseline specification is:

$$\Delta \log p_{fpot} = \sum_{\tau=0}^8 \alpha_{\tau} \Delta \log e_{ot-\tau} + \beta X_{fpot} + \varepsilon_{fpot}, \quad (22)$$

where $\Delta \log e_{ot}$ is the quarterly change in bilateral exchange rates, X_{fpot} includes a set of controls and fixed effects, and α_0 is the parameter of interest capturing the short-run exchange rate pass-through on import prices at the border. Following [Burstein and Gopinath \(2014\)](#), I include the inflation rate and GDP growth in the origin country and its 8 lags as control. I include importer-product-origin and time fixed effects.

In line with estimates from the literature, the main specification yields an average pass-through rate of 66 percentage points in the full sample (Column (1) in Table 6 in Appendix B). Using alternative sets of fixed effects or abstracting from control variables, I obtain an average pass-through rate ranging from 59 to 68 percentage points. I estimate Equation (22) at the HS2 level and calibrate the vector Ψ^F in Equation (17) after constructing a crosswalk between HS2 codes and product categories in the input-output tables.¹⁶ Substantial heterogeneity is present across product categories, with most of the distribution ranging from 0.4 to 0.8 (right panel of Figure 5).

3 Quantitative Results

This section quantifies the importance of domestic wedges and border pass-through, as well as their heterogeneity, for the response of domestic prices and the CPI to exchange rate fluctuations. I provide two baseline scenarios to gauge the role of wedges and to evaluate the performance of the quantitative model.

Empirical and Neoclassical Baseline First, following [Burstein and Gopinath \(2014\)](#), I estimate the average CPI response to a 1 p.p. depreciation of the Chilean peso during 1990–2019 using the specification:

$$\Delta \log CPI_t = \sum_{\tau=0}^8 \beta_{\tau} \Delta \log e_{t-\tau} + \sum_{\tau=0}^8 \gamma_{\tau} X_{t-\tau} + \varepsilon_t, \quad (23)$$

where CPI is the Chilean consumer price index at quarterly frequency, e is the trade-weighted nominal exchange rate between the peso and exporters' currencies, and X is a vector of controls including trading partners' inflation as a proxy for foreign costs, and Chile's real GDP growth ([Campa and Goldberg, 2005](#); [Burstein and Gopinath, 2014](#)). I include up to eight lags to control for autocorrelation in inflation, exchange rates, and controls.

The estimation delivers a CPI sensitivity of about 8.2 percent, consistent with existing findings. The estimate is robust to alternative lag structures and exchange rate definitions.

¹⁶ For product categories without a match in the HS classification (e.g., services), I calibrate the entries in Ψ^F to the average pass-through rate estimate obtained from the customs data.

Matching this magnitude provides a useful benchmark against which to evaluate the quantitative framework in Section 1.1. Further details on estimation and robustness are reported in Appendix B.4.

Second, I construct a counterfactual “neoclassical” benchmark in which wedges are absent and pass-through is complete at all stages. Formally, I set $\eta^F = \Gamma^D = \Delta^D = \Phi^D = \mathbf{1}$ and $\Lambda^F = \Lambda^D = 0$. In this economy, the CPI sensitivity depends solely on the import content of consumption and production, adjusted for domestic input–output linkages. Comparing the fully calibrated model with this frictionless benchmark highlights the quantitative role of wedges.

3.1 The role of domestic wedges

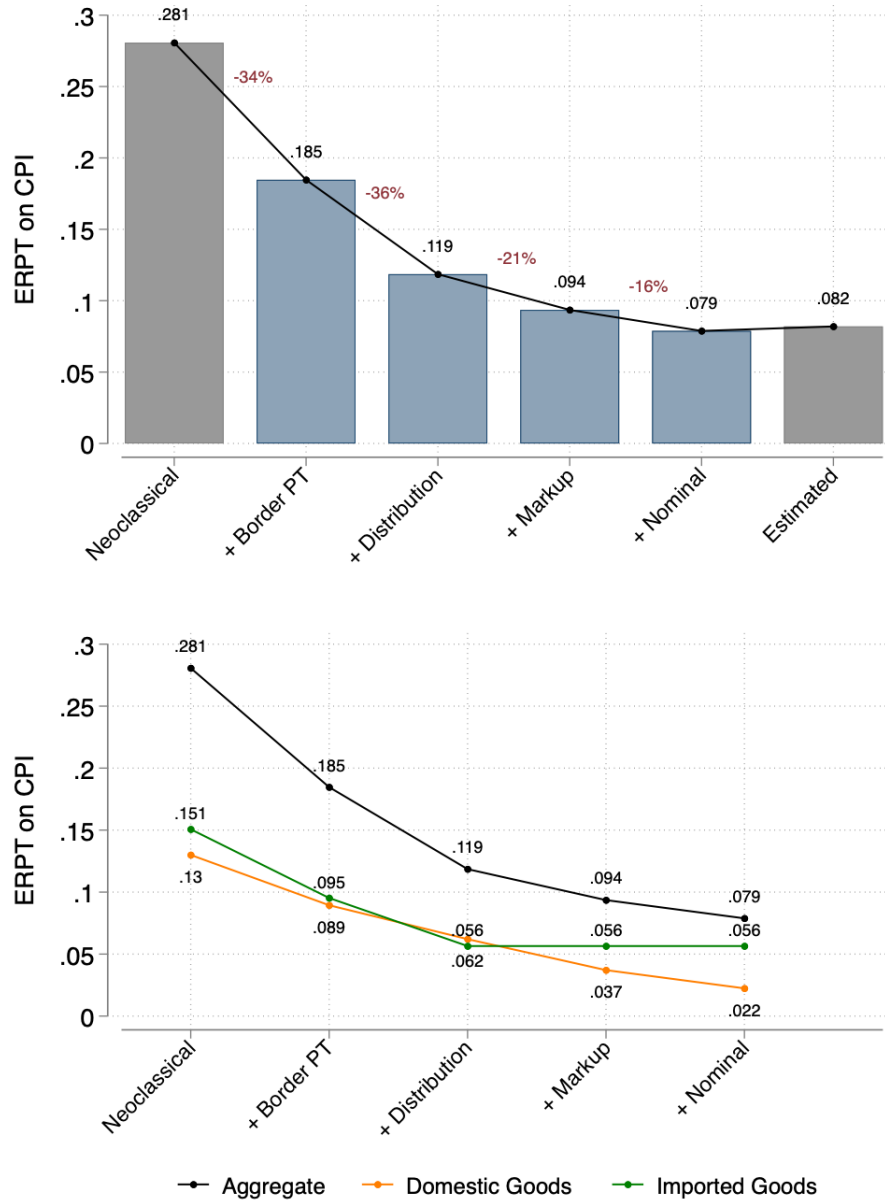
The benchmark model, which embeds both domestic wedges and incomplete border pass-through, closely replicates the estimated CPI sensitivity: 7.9 percent in the model versus 8.2 percent in the data (second-last and last bars in the top panel of Figure 6, respectively). This close alignment confirms that the simple measurement equations introduced in Section 1 capture the relevant transmission mechanisms without sacrificing quantitative accuracy.

The low CPI sensitivity is primarily driven by domestic wedges – mechanisms that operate independently of border prices. In a neoclassical scenario with complete pass-through, CPI sensitivity would be almost four times larger than observed, at 28 percent (first bar on the top panel of Figure 6). Introducing incomplete pass-through at the border dampens the response only partially relative to the neoclassical scenario, from 28 to 18.5 percent (second bar on the top panel of Figure 6). Border price sensitivity dampens CPI sensitivity but cannot explain the low observed value. The addition of domestic wedges brings CPI sensitivity down to observed levels. CPI sensitivity decreases by an additional 57%, from 18.5 to 7.9 percent, indicating that domestic factors play a larger role in moderating CPI sensitivity relative to border price sensitivity. This shows that incomplete pass-through at the border and domestic frictions are complementary as neither mechanism alone can account for the muted response of consumer prices to exchange rate fluctuations.

All domestic wedges individually contribute to the low sensitivity of CPI, mainly through their effects on domestically produced goods. CPI sensitivity decreases by 36% when distribution costs are introduced. By contrast, the decrease is more moderate when variable markups and nominal rigidities are introduced (21 and 16 percent, respectively), resonating the evidence in Nakamura and Zerom (2010). The reason is that markups and rigidities operate narrowly through domestic goods produced with imported inputs, whereas distribution costs apply economy-wide, shaping the prices of both domestic and imported final goods. However, when focusing on the relative changes for domestic and imported final goods separately (orange and green lines in the bottom panel of Figure 6, respectively), nominal rigidities and variable markups contribute more than distribution costs to reducing the sensitivity of domestic final goods.¹⁷

¹⁷ Figure 16 in Appendix C provides qualitatively similar results from excluding one wedge - distribution costs, variable markups, nominal rigidities, or border price responses - at a time from the benchmark

Figure 6: Domestic Wedges vs Border Price Dynamics



Note: The top panel plots the aggregate CPI sensitivity to a one percent depreciation in the exchange rate across different scenarios. The first bar on the left represents the average CPI sensitivity in a counterfactual frictionless, neoclassical economy where pass-through is complete and domestic wedges are absent. The blue bars in the middle sequentially incorporate incomplete pass-through into import prices at the border, distribution costs, variable markups, and nominal rigidities to the neoclassical scenario. The red numbers between the bars indicate the percentage reduction in CPI sensitivity attributable to each additional element. The final bar on the right, labeled "Estimated", displays the CPI sensitivity to exchange rate fluctuations estimated at the quarterly level over the period 1990–2019 (see Appendix B.4 for estimation details). On the bottom panel, the black line represents the aggregate CPI sensitivity to a one percent depreciation in the exchange rate. The green and orange lines represent the decomposition into imported final consumption ("Imported Goods"), i.e. direct exposure, and domestic final consumption ("Domestic Goods"), i.e. indirect exposure, respectively. Values represent the average for the years 2008–2019.

3.2 Heterogeneity and composition effects

Focusing only on the aggregate sensitivity of CPI may obscure substantial heterogeneity across products. This heterogeneity arises from various dimensions, such as domestic wedges, exposure to import and consumption shares, and their interaction with one another.

The top panel of Figure 7 documents the significant dispersion in product-level sensitivities to exchange rate fluctuations. For domestic final consumption goods, sensitivities range from 0 to 0.3, while for imported final consumption goods, they span from 0 to 0.85. Figure 17 in Appendix C shows that sectoral heterogeneity aligns with economic intuition. Service sectors exhibit low sensitivity to exchange rates given their relatively low exposure to imported inputs, whereas chemical and metal products display higher sensitivities due to greater dependence on imported inputs. This pronounced heterogeneity shapes the aggregate CPI sensitivity by interacting with the consumption shares of individual products. Simultaneously, it reflects the interplay of product-level border price dynamics, import exposure in production, and domestic wedges.

Heterogeneity in Wedges I find that the heterogeneity in wedges dampens the CPI sensitivity relative to the benchmark model. To quantify the impact of heterogeneity, I set each wedge to its average value across products and compare the resulting sensitivities with those of the benchmark model. In this scenario, labeled "Joint" in the bottom panel of Figure 7, the sensitivity of CPI increases by 55 percent relative to the benchmark model. Interestingly, the effects of heterogeneity in wedges differ in magnitude and direction between domestic and imported final consumption goods. When heterogeneity in wedges is abstracted from, imported goods exhibit lower sensitivity, whereas domestic goods display increased sensitivity, highlighting the differential impact of wedges across product types.

While joint heterogeneity across all wedges dampens CPI sensitivity, heterogeneity in individual wedges may amplify CPI sensitivity. For each wedge, I assess the impact of its heterogeneity by setting it equal to its average value across products while holding all other factors at benchmark levels. Each scenario is represented in the bottom panel of Figure 7.¹⁸ Accounting for the heterogeneity in distribution costs and border price dynamics reduces CPI sensitivity by approximately 25 percent, driving the results from the joint heterogeneity scenario. In contrast, heterogeneity in nominal rigidities amplifies CPI sensitivity by approximately 7 percent; heterogeneity in markups also amplifies CPI sensitivity, but its effect is only marginal (around 1 percent), as expected from the small calibrated covariances shown in Figure 4.

The effects of heterogeneity in individual wedges differ in magnitude and direction between domestic and imported final consumption goods. For instance, heterogeneity in distribution costs amplifies CPI sensitivity for domestic goods while dampening it for imported goods. Heterogeneity in border price sensitivity has a qualitatively similar effect on both domestic and

model.

¹⁸ The absence of heterogeneity in variable markups is defined as i) no within-sector heterogeneity (covariance terms set to zero) and ii) the same average markup elasticity across products, which is set equal to the average across products.

imported goods but is quantitatively stronger for imported goods, increasing CPI sensitivity by 30 percent compared to 15 percent for imported goods. By definition, nominal rigidities and variable markups affect only the sensitivity of domestic goods.

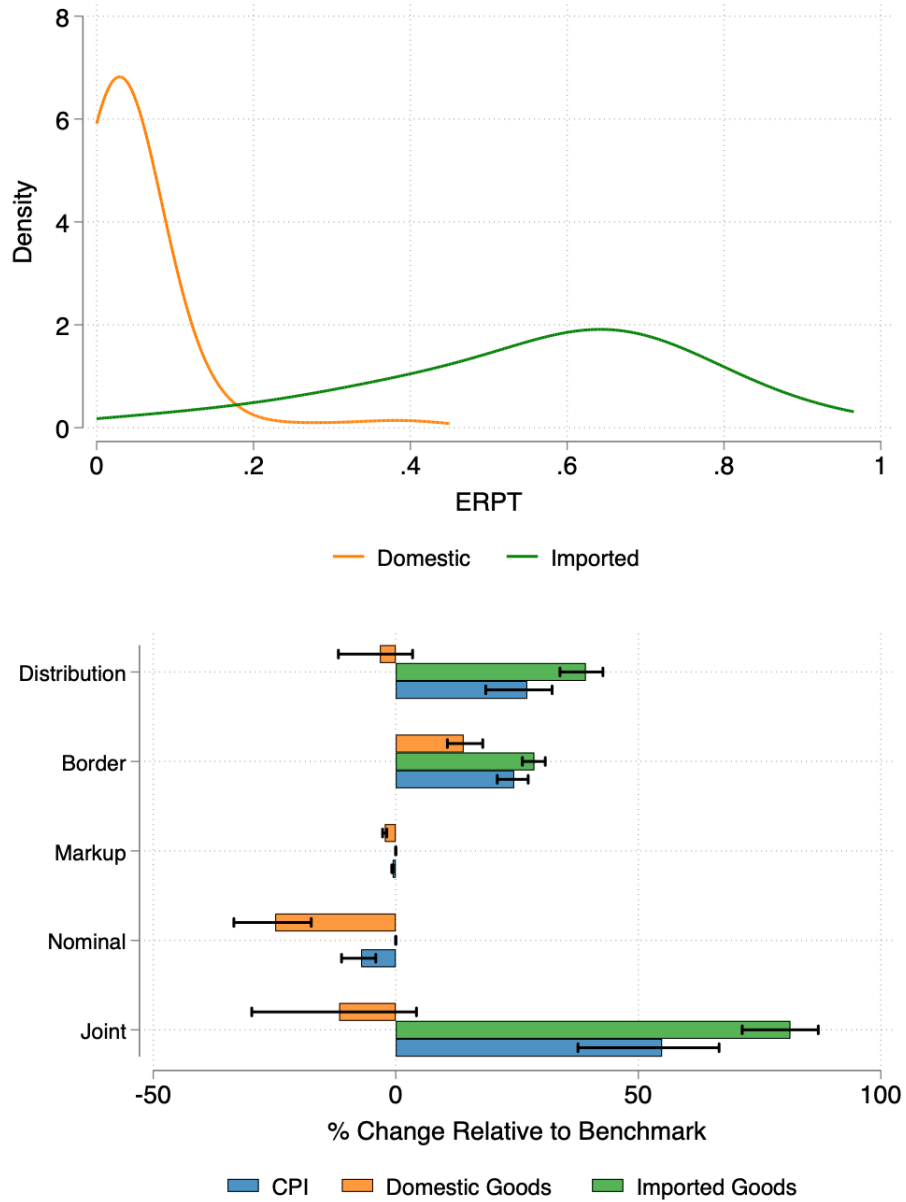
Within Sector Heterogeneity The contribution of markups heterogeneity encompasses the role of both within-product heterogeneity across firms, and the heterogeneity across products. Figure 18 in Appendix C shows that both margins are qualitatively and quantitatively similar in driving the overall effect of markups heterogeneity. Accounting for heterogeneity in markups amplifies the response of domestic goods to exchange rate fluctuations by 2.25 percent. Approximately 60 percent of the total effect (1.4 percent) is due to heterogeneity across sectors, while the remaining portion (0.85 percent) is due to heterogeneity within sectors. Moreover, the effect of within-sector heterogeneity is equally driven by the heterogeneity in the use of domestic and imported inputs across firms.

Composition effect I find that heterogeneity in consumption shares further dampens CPI sensitivity, as expenditure is concentrated in goods less exposed to imports. I compute the sensitivity of CPI when consumption shares are set equal to the average consumption share across product without affecting the extensive margin, i.e. without altering the set of products consumed. Figure 19 in Appendix C shows that the heterogeneity in consumption shares dampens CPI sensitivity by approximately 70 percent, with its effects being stronger for imported final goods relative to domestic goods (80 and 50 percent, respectively). Thus, exchange rate fluctuations might ultimately influence domestic redistribution as consumers rely on varying mixes of domestic and imported goods (Cravino and Levchenko, 2017a; Jaravel, 2021).

Correlation in Primitives of the Economy I provide evidence consistent with the idea that the impact of heterogeneity on CPI sensitivity depends on how wedges correlate with the use of imported goods in production and consumption. Panel A of Table 1 shows that the share of imported inputs used in the production of domestic goods, which measures the exposure of domestic goods to exchange rate fluctuations, is positively correlated with the frequency of price adjustment and the incomplete pass-through rate due to variable markups in the sector (Columns (1) and (2)). Thus, sectors more exposed to imports are also those with stronger adjustment, reinforcing their role in aggregate sensitivity. Moreover, the share of imported inputs is negatively correlated with border price sensitivity of imported inputs (Column (3)). This finding aligns with the observation that abstracting from heterogeneity in border price sensitivity amplifies CPI sensitivity, as the most relevant imported inputs are also those with lower sensitivity to exchange rate fluctuations.

Panel B of Table 1 shows that the share of imported inputs used in the production of domestic goods is negatively correlated with the share of distribution cost in retail prices (Columns (3) and (4)) and the consumption share (Column (1)). Therefore, sectors with higher distribution costs or larger consumption shares are less exposed to imports, explaining why removing these heterogeneities increases sensitivity. Moreover, the negative correlation between the share of

Figure 7: On Heterogeneity



Note: The top panel displays the distribution of product-level sensitivities for final goods in the consumption basket. The orange line represents domestically produced goods, while the green line represents imported final goods. The analysis uses data from 2008 to 2019. The bottom panel presents the percentage change in CPI sensitivity relative to the benchmark economy under various scenarios. The label of each scenario indicates the dimension of heterogeneity that is dropped from the analysis. When abstracting from heterogeneity, the relevant dimension is set equal to the average across products. The "Joint" case indicates the scenario in which all dimensions of heterogeneity are removed. The blue bars show the change in CPI sensitivity, while the orange and green bars show the percentage change in the sensitivity of domestic and imported final goods, respectively, relative to the benchmark economy. The bands represent the range of estimated CPI sensitivities for the years 2008 - 2019.

Table 1: Correlation between Dimensions of Heterogeneity

Panel A	Nominal	Border ERPT	Average Markup	Covariance Markup-Domestic	Covariance Markup-Foreign
	(1)	(2)	(3)	(4)	(5)
Share Imported Inputs	0.155 (0.0419)	-0.045 (0.0231)	0.054 (0.0097)	0.117 (0.0003)	-0.061 (0.0008)
<i>N</i>	2135	2135	2135	2135	2135
Panel B	Consumption Share Domestic Goods	Consumption Share Imported Goods	Distribution Domestic Goods	Distribution Imported Goods	
	(1)	(2)	(3)	(4)	
Share Imported Inputs	-0.0142 (0.0020)		-0.1636 (0.0295)	-0.2546 (0.0302)	
Border ERPT		-0.0015 (0.0004)			
<i>N</i>	2135	2135	2135	2135	

Note: Panel A reports the regression coefficients of the aggregate share of imported inputs used in production on various sector-level primitives. These include the frequency of price adjustment (labeled "Nominal"), the average border price sensitivity of imported inputs, the average pass-through rate, and the covariance between pass-through rates and domestic or imported goods. Panel B presents the regression coefficients of the aggregate share of imported inputs used in production on the consumption share of domestic goods in the final consumption basket and the distribution margin. Additionally, it includes the regression coefficient of border price sensitivity on the consumption share of imported final goods. Standard errors are reported in parentheses, and all coefficients are standardized to allow for comparison of the variance explained.

imported final goods and the border price sensitivity aligns with the increased sensitivity of imported final goods when heterogeneity in consumption shares is removed (Column (2)).

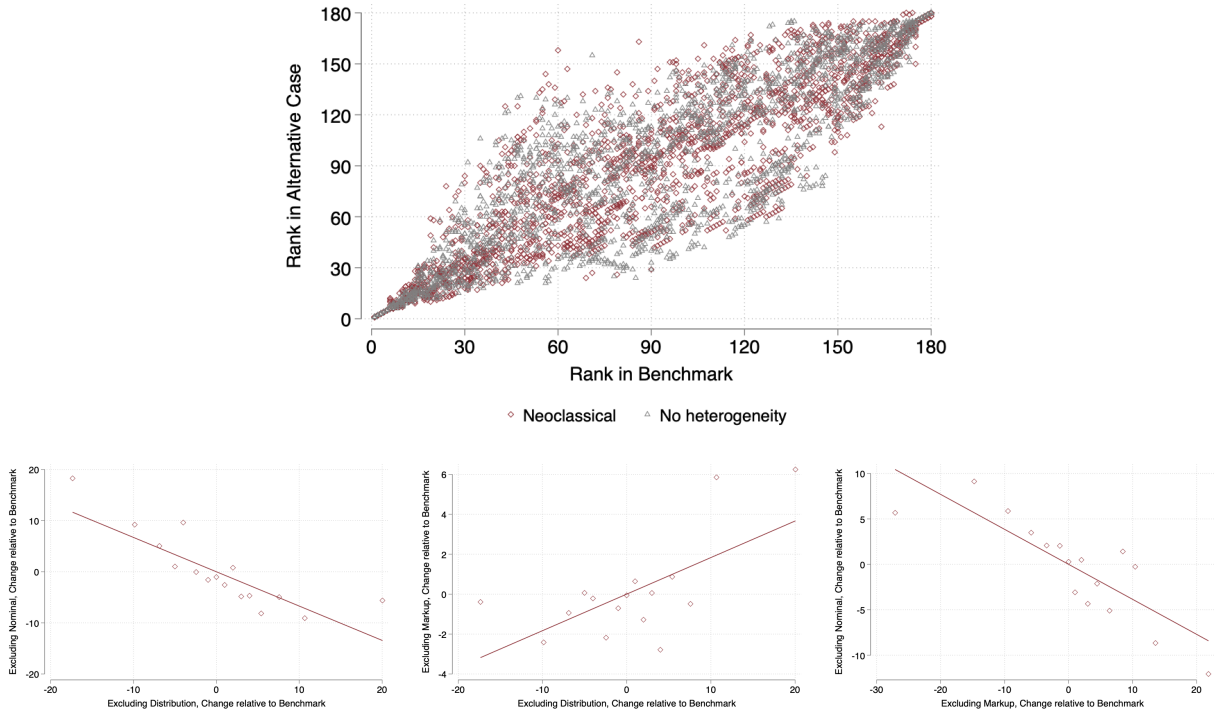
Taken together, the results in Sections 3.1 and 3.2 show that the muted aggregate response of the CPI cannot be understood without domestic wedges or without accounting for their rich heterogeneity across products. While average effects of domestic wedges depress sensitivity, their heterogeneity across sectors often dampens it further by concentrating the largest frictions in precisely those goods most exposed to imports. The composition of consumption adds another layer of attenuation, as larger expenditure shares tend to fall on products less exposed to import costs. These findings underscore that the muted aggregate response of CPI reflects not only the direct role of wedges, but also the way in which economic structure channels exchange rate shocks unevenly across products and consumers.

3.3 Identity effect

Incorporating wedges and their heterogeneity alters which products matter most for CPI sensitivity, a phenomenon often referred to as the identity effect.

Direct and Indirect Exposure The bottom panel of Figure 6 shows that the fully calibrated economy predicts that the majority of CPI sensitivity arises from the consumption of imported final goods, corresponding to the direct exposure in Equation (6). Ignoring wedges leads to overstating the role of domestic final goods, i.e., indirect exposure, since wedges disproportionately mute their response. In fact, in a frictionless neoclassical economy, direct and

Figure 8: Micro Identity Effects



Note: The top panel compares the ranking of products contributing the most to CPI sensitivity in the benchmark model (x-axis) to their ranking in alternative scenarios (y-axis). The alternative scenarios include: a frictionless neoclassical economy that excludes all domestic wedges and border price sensitivity (red diamonds) and an economy that omits the heterogeneity in domestic wedges and border price sensitivity (gray triangles). The three bottom panels compare the changes in rankings relative to the benchmark economy when omitting specific combinations of domestic wedges: nominal rigidities and distribution costs (left panel), variable markups and distribution costs (middle panel), and nominal rigidities and variable markups (right panel). Solid lines indicate a linear fit within the binscatter plot.

indirect import exposure contribute almost equally to overall price changes (15 and 13 percent, respectively). Introducing incomplete pass-through at the border does not alter the relative importance of the two types of exposure. However, the relative importance shifts markedly when domestic wedges are incorporated – particularly nominal rigidities and variable markups – as these factors exclusively influence the sensitivity of domestically produced goods. Standard practices, which often fail to account for domestic wedges, quantify direct exposure using frameworks akin to the frictionless economy, leading to potential mischaracterizations of CPI sensitivity attributing equal importance to direct and indirect exposure ([Goldberg and Campa, 2010](#); [Burstein et al., 2003](#); [Gopinath, 2015](#)).

Identity Effect at the Product Level Which products matter most for CPI sensitivity shifts once (heterogeneous) wedges are incorporated. The top panel of Figure 8 shows how the ranking of the products contributing most to overall CPI sensitivity changes depending on the inclusion of domestic wedges in the economy. Relative to a frictionless neoclassical benchmark or an economy without wedge heterogeneity, product rankings shift significantly, with some products moving as many as 80 positions. The three bottom panels in Figure 8 show that these changes in rankings are specific to the domestic wedge being omitted. In three

separate scenarios, I exclude one domestic wedge at a time and calculate the change in product rankings relative to the benchmark economy. Omitting nominal rigidities generates identity effects that contrast sharply with those observed when distribution costs or variable markups are omitted. The identity effects arising from distribution costs and variable markups show instead positive correlation. These findings underscore the importance of carefully considering the specific dimensions of heterogeneity at play when identifying the products most critical to the transmission of exchange rate fluctuations.

3.4 On the Role of Input-Output linkages

The sensitivity of domestically produced goods arises not only from the direct use of imported inputs in production but also from domestic input-output linkages. In other words, even when a domestically produced good does not directly use imported intermediate inputs, the domestic inputs in its production process might themselves be exposed to imports.

I show that the contribution of input-output linkages to CPI sensitivity depends critically on the presence and heterogeneity of wedges. Figure 9 compares the CPI sensitivity under two scenarios: one with roundabout production (solid line) and one without it (dashed line). In a neoclassical economy, input-output linkages amplify CPI sensitivity by 74%. When domestic wedges are incorporated, the amplification effect of input-output linkages diminishes to just 25% in the benchmark model (Basu, 1994; Pasten et al., 2020).¹⁹ This reduction occurs because domestic wedges dampen price sensitivity and limit downstream propagation at each stage of the production network (Carvalho and Tahbaz-Salehi, 2019).²⁰ Incomplete pass-through at border does not affect the role of input-output linkages because it does not directly interact with the domestic Leontief matrix.

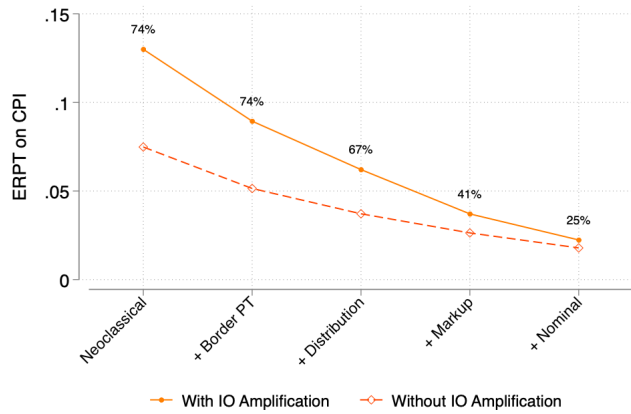
The inclusion of domestic wedges also alters the centrality of products within the domestic input-output network and their correlation with the use of imported inputs. To assess this, I use the PageRank centrality measure, which evaluates the relative importance of each product in the network. Figure 10 shows how product centrality changes when edge weights are based solely on input shares from input-output tables versus when these weights are adjusted to account for the presence of domestic wedges. Accounting for wedges flattens the relationship between product centrality and exposure to imported inputs, thereby reducing the overall sensitivity of CPI to exchange rate fluctuations.²¹

¹⁹ Figure 21 in Appendix C shows that omitting a domestic wedge from the fully calibrated model increases the sensitivity of CPI by approximately three times more when input-output linkages are present in the economy. This effect is quantitatively similar across the three domestic wedges considered – distribution costs, nominal rigidities, and variable markups.

²⁰ Formally, the adjusted Leontief inverse matrix captures both direct and indirect downstream propagation. Abstracting from domestic wedges involves using the Leontief inverse matrix instead of the adjusted one in Equation (17), with the former leading to stronger amplification.

²¹ Figure 23 in Appendix C shows that these qualitatively patterns holds when product centrality is constructed using the average between indegree and outdegree centrality of each product in the domestic input-output network.

Figure 9: Effects of Input-Output Linkages



Note: The Figure plots the CPI sensitivity to a one percent depreciation in the exchange rate across different scenarios. The first point on the left represents the average sensitivity in a counterfactual frictionless, neoclassical economy where pass-through is complete and domestic wedges are absent. Each subsequent point sequentially incorporates incomplete pass-through into import prices at the border, distribution costs, variable markups, and nominal rigidities to the neoclassical scenario. The solid line depicts the aggregate CPI sensitivity to a one percent depreciation in the exchange rate, accounting for the amplification effects of domestic input-output linkages. In contrast, the dashed line represents the CPI sensitivity when the amplification from domestic input-output linkages is omitted.

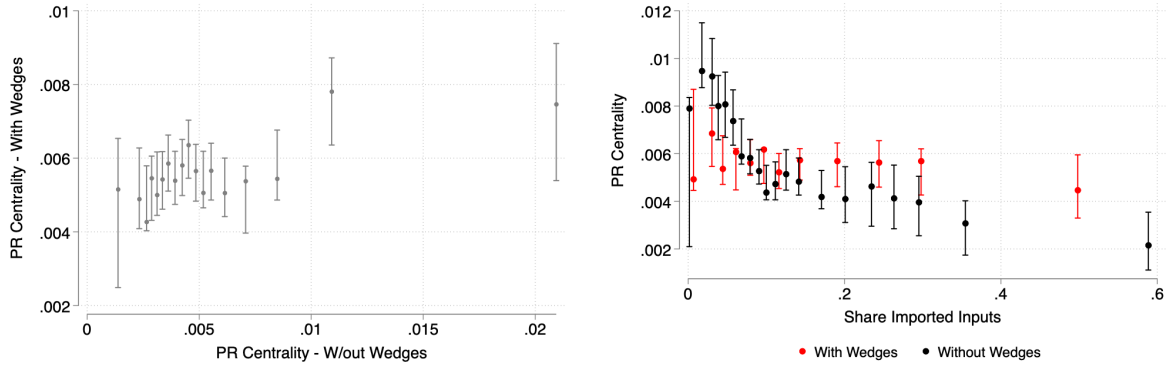
Taken together, the results in Sections 3.3 and 3.4 suggest that, once domestic wedges and their heterogeneity are accounted for, it is imported consumption goods—not domestic input-output linkages—that drive CPI sensitivity. In practice, this means that shocks matter most where households directly consume foreign goods, while the amplification role of production linkages is muted by the presence of wedges. The composition and distribution of wedges across products therefore shape not only the magnitude of pass-through, but also which sectors emerge as critical for the aggregate response.

3.5 Robustness

Figure 22 in Appendix C illustrates the stability of CPI sensitivity over the period 2008–2019, suggesting that the average effect reported in Figure 6 does not obscure significant time variation. The only notable deviation occurs in 2009, where the sensitivity of the CPI decreases by 1.5 p.p.. The decomposition into domestic and imported final goods reveals that this decline is entirely attributable to imported goods, consistent with the effects of the great trade collapse during the global financial crisis.

Table 8 in Appendix C evaluates the robustness of the benchmark economy’s calibration. Following Edmond et al. (2018), I construct the average pass-through rate and the covariance terms arising from variable markups (Γ^D , Λ^D , and Λ^F) using cost shares instead of revenue shares. Figure 14 in Appendix B demonstrates that these terms, constructed using cost shares, exhibit near-perfect correlation with their benchmark counterparts based on revenue shares. As expected, the sensitivities of CPI, domestic goods, and imported final goods remain consistent with those in the benchmark model.

Figure 10: Product Centrality and Imported Inputs



Note: The left panel illustrates the relationship between the PageRank centrality of a product in the input-output network when the edges are constructed based on input-output tables that include domestic wedges (y-axis) versus those that omit domestic wedges (x-axis). The right panel depicts the relationship between a product’s centrality in the domestic production network and the share of imported inputs used in its production. Centrality is measured using the PageRank centrality measure, with edges weighted by input-output linkages that either include domestic wedges (red dots) or omit domestic wedges (black dots). Both panels use data spanning the years 2008 to 2019. Both panels use binscatters constructed while absorbing product and year fixed effects. The figures include 95% confidence intervals.

The frequency of price adjustment is calibrated using data from the BLS. To account for potential differences between the US and Chile, I rescale the frequency distribution to match an average monthly frequency of price adjustment of 30%, as estimated by [Aruoba et al. \(2022\)](#) using confidential daily transaction data from the Chilean Tax Authority. This adjustment implies an average quarterly adjustment probability of 0.65, slightly higher than the benchmark calibration. Consequently, the effect of nominal rigidities on exchange rate pass-through into CPI increases marginally from 0.079 to 0.082. Nevertheless, the benchmark model’s overall performance is robust to this calibration adjustment.

The customs data used to calibrate border import price sensitivity exclude service imports. For unmatched product categories in the HS classification, I assume their sensitivity equals the average pass-through rate estimated from the customs data. To test this assumption, I explore two extreme cases: full pass-through for imported services at the border and zero pass-through. Table 8 shows that complete pass-through increases CPI sensitivity marginally to 0.084, while zero pass-through reduces it by nearly 1 p.p.. Most of the impact occurs through the sensitivity of domestic final goods, indicating that imported services are primarily used as intermediate inputs rather than final goods. Overall, the benchmark model’s performance remains largely unaffected by this assumption.

4 Conclusion

In this paper, I have examined the role of domestic wedges and their heterogeneity for the (in)sensitivity of domestic prices to exchange rate fluctuations. I find that domestic wedges – distribution costs, variable markups, and nominal rigidities – account for most of the low sensitivity of the CPI to exchange rate movements. Additionally, I show that accounting for the

presence and heterogeneity of domestic wedges has important implications for identifying the products or product groups most relevant to overall CPI sensitivity. Differently from results based on standard strategies, I find that most of the sensitivity arises from direct exposure (imported final consumption), as domestic wedges dampen the response of domestically produced goods (indirect exposure) to a greater extent. Furthermore, the identity of the products contributing most to the transmission of exchange rate fluctuations depends on the specific sources of heterogeneity considered. This underscores the importance of jointly accounting for the wedges and mechanisms included in the analysis. In this regard, the model and calibration strategy used in this paper can help guide future research on the relationship between domestic prices and exchange rates.

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A Appendix - Theory

A.1 Derivations of key pass-through equations

Producers of sectoral varieties. Consider a domestic sector i . The N_i firms indexed by k within the sector employ a CRS technology to produce a variety of the same product:

$$y_{ik} = F_i(l_{i,k}, x_{i1,k}, \dots, x_{iN,k}) = \zeta_{ik} l_{i,k}^{\alpha_{ik,l}} \prod_{j=1}^N x_{ij,k}^{\alpha_{ik,j}} \quad \text{with } \alpha_{ik,l} + \sum_{j=1}^N \alpha_{ik,j} = 1 \quad \forall k, \quad (24)$$

where each firm k potentially uses a different mix of inputs (i.e. exponents are k -specific).

Given the assumption on the distributor's aggregating technology, producers charge a variable markup over the marginal cost:

$$\widetilde{p}_{ik} = \mu_{ik} mc_{ik} \quad \text{with } mc_{ik} \propto w^{\alpha_{ik,l}} \prod_{j=1}^N p_j^{\alpha_{ik,j}}, \quad (25)$$

where $\widetilde{p}_{ik}$ is the price paid by the distributor for variety k , $\mu_{ik} = \frac{\varepsilon_{ik}}{\varepsilon_{ik}-1}$ is the markup charged, that depends on the perceived demand elasticity ε_{ik} , and the expression for the marginal cost, mc_{ik} , comes from the specific production function assumed.

Derivation of sectoral price inflation We focus on the direct effect of a change in the marginal cost on the producer price at fixed aggregate price (Burstein and Gopinath, 2014). Consider a domestic sector $i \in H$. At the firm level, the change in the price of variety k in sector i is given by:

$$\partial \log \widetilde{p}_{ik} = \kappa_{ik} \partial \log mc_{ik},$$

where $\kappa_{ik} \equiv \frac{1}{1+\Gamma_{ik}}$, and $\Gamma_{ik} \equiv -\frac{\partial \log \mu_{ik}}{\partial \log \widetilde{p}_{ik}}$ is the markup elasticity.

It follows that the sectoral price inflation is:

$$\partial \log \widetilde{p}_i \equiv \sum_{k \in N_i} S_{ik} \partial \log \widetilde{p}_{ik} = \sum_{k \in N_i} S_{ik} \kappa_{ik} \partial \log mc_{ik} = \bar{\kappa}_i \overline{\partial \log mc_i} + cov_{S_{ik}}(\kappa_{ik}, \partial \log mc_{ik}), \quad (26)$$

where S_{ik} is the revenues share of firm k in sector i ; $\bar{x}$ is the revenues-weighted average of variable x , i.e. $\bar{x} = \sum_{k \in N_i} S_{ik} x_{ik}$; and $cov_{z_{ik}}(x_{ik}, y_{ik})$ is the z -weighted covariance between two variables x and y , i.e. $cov_{z_{ik}}(x_{ik}, y_{ik}) = \sum_{k \in N_i} z_{ik} (x_{ik} - \bar{x})(y_{ik} - \bar{y})$.

Derivation of marginal cost changes Given the production technology in Equations (24) and (25), the change in the marginal cost of producer k in domestic sector i following a change in input prices is as follows:

$$\partial \log mc_{ik} = \sum_{j=1}^N \alpha_{ik,j} \partial \log p_j, \quad (27)$$

where $\alpha_{ik,j}$ is the exposure of firm k in sector i to input j , and p_j is the retail price of input j . It follows that we can write:

$$\overline{\partial \log mc_i} = \sum_{k \in N_i} S_{ik} \sum_{j=1}^N \alpha_{ik,j} \partial \log p_j = \sum_{j=1}^N \bar{\alpha}_{ij} \partial \log p_j, \quad (28)$$

where $\bar{\alpha}_{ij} = \sum_{k \in N_i} S_{ik} \alpha_{ik,j}$ is the revenue-weighted average input share of product j in sector i . Moreover, we can rewrite the covariance term in Equation (26) as follows:

$$cov_{S_{ik}}(\kappa_{ik}, \partial \log mc_{ik}) = \sum_j^N \partial \log p_j cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j}). \quad (29)$$

Derivation of retail price inflation Given the assumption of the distribution sector and the Calvo-style nominal rigidities, we can write the change in the retail price of a *domestic* good i as follow:

$$\partial \log p_i = \phi_i \delta_i \partial \log \tilde{p}_i. \quad (30)$$

Substituting Equations (26), (28), and (29), we obtain:

$$\partial \log p_i = \phi_i \delta_i \sum_j^N \partial \log p_j [\bar{\alpha}_{ij} \bar{\kappa}_i + cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j})]. \quad (31)$$

Part of the inputs used in the production of domestic goods are imported, i.e. N can be partitioned into domestic (n_D) and imported inputs (n_F), i.e. $N = n_D + n_F$. We can therefore rewrite Equation (31) highlighting this difference:

$$\begin{aligned} \partial \log p_i = & \phi_i \delta_i \sum_j^{n_D} \partial \log p_j [\bar{\alpha}_{ij} \bar{\kappa}_i + cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j})] + \\ & \phi_i \delta_i \sum_j^{n_F} \partial \log p_j [\bar{\alpha}_{ij} \bar{\kappa}_i + cov_{S_{ik}}(\kappa_{ik}, \alpha_{ik,j})]. \end{aligned}$$

We can therefore solve for the change in the retail price of a domestic good $i \in n_D$. In matrix form, we can write:

$$\Pi^D = [\mathbf{I} - \Phi^D \Delta^D (\Gamma^D \mathbf{S}^D + \Lambda^D)]^{-1} \Phi^D \Delta^D (\Gamma^D \mathbf{S}^F + \Lambda^F) \Pi^F, \quad (32)$$

with:

$$\begin{aligned}
\Phi^D &= \begin{bmatrix} \phi_1 & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \phi_{n_D} \end{bmatrix}, & \Delta^D &= \begin{bmatrix} \delta_1 & 0 & 0 \\ \vdots & \ddots & \vdots \\ 0 & 0 & \delta_{n_D} \end{bmatrix}, & \Gamma^D &= \begin{bmatrix} \bar{\kappa}_1 & \cdots & \bar{\kappa}_{n_D} \\ \vdots & \ddots & \vdots \\ \bar{\kappa}_1 & \cdots & \bar{\kappa}_{n_D} \end{bmatrix}, \\
\Lambda^F &= \begin{bmatrix} cov_{S_{1k}}(\kappa_{1k}, \alpha_{1k,1}) & \cdots & cov_{S_{1k}}(\kappa_{1k}, \alpha_{1k,n_F}) \\ \vdots & \ddots & \vdots \\ cov_{S_{n_D k}}(\kappa_{n_D k}, \alpha_{n_D k,1}) & \cdots & cov_{S_{n_D k}}(\kappa_{n_D k}, \alpha_{n_D k,n_F}) \end{bmatrix}, & \mathbf{S}^F &= \begin{bmatrix} \bar{\alpha}_{11} & \cdots & \bar{\alpha}_{1n_F} \\ \vdots & \ddots & \vdots \\ \bar{\alpha}_{n_D 1} & \cdots & \bar{\alpha}_{n_D n_F} \end{bmatrix}, \\
\Lambda^D &= \begin{bmatrix} cov_{S_{1k}}(\kappa_{1k}, \alpha_{1k,1}) & \cdots & cov_{S_{1k}}(\kappa_{1k}, \alpha_{1k,n_D}) \\ \vdots & \ddots & \vdots \\ cov_{S_{n_D k}}(\kappa_{n_D k}, \alpha_{n_D k,1}) & \cdots & cov_{S_{n_D k}}(\kappa_{n_D k}, \alpha_{n_D k,n_D}) \end{bmatrix}, & \mathbf{S}^D &= \begin{bmatrix} \bar{\alpha}_{11} & \cdots & \bar{\alpha}_{1n_D} \\ \vdots & \ddots & \vdots \\ \bar{\alpha}_{n_D 1} & \cdots & \bar{\alpha}_{n_D n_D} \end{bmatrix}, \\
\Pi^D &= [\partial \log p_1, \dots, \partial \log p_{n_D}]^T, & \Pi^F &= [\partial \log p_1, \dots, \partial \log p_{n_F}]^T.
\end{aligned}$$

where Π^D and Π^F are the $n_D \times 1$ and $n_F \times 1$ vectors of changes in domestic and foreign prices, respectively; Φ^D and Δ^D are the $n_D \times n_D$ diagonal matrices of sectoral distribution margins and Calvo parameters, respectively; $\mathbf{S}^D$ ($\mathbf{S}^F$) is the $n_D \times n_D$ ($n_D \times n_F$) matrix of average use of domestic (foreign) inputs in the production of domestic goods; Γ^D is the $n_D \times n_D$ matrix of sectoral revenue-weighted average pass-through; Λ^D (Λ^F) is the $n_D \times n_D$ ($n_D \times n_F$) matrix of sectoral covariances between firm-level markup elasticity and the use of domestic (imported) inputs in the production of domestic goods.

Derivation of exchange rate pass-through into domestic inflation Equation (17) in the main text follows from Equation (32). The pass-through into the retail price of domestic goods, $\eta^D = \left[\frac{\partial \log p_1}{\partial \log e}, \dots, \frac{\partial \log p_H}{\partial \log e} \right]^T$ can be written as:

$$\begin{aligned}
\eta^D &= [\mathbf{I} - \Phi^D \Delta^D (\Gamma^D \mathbf{S}^D + \Lambda^D)]^{-1} \Phi^D \Delta^D (\Gamma^D \mathbf{S}^F + \Lambda^F) \eta^F \\
&= [\mathbf{I} - \Phi^D \Delta^D (\Gamma^D \mathbf{S}^D + \Lambda^D)]^{-1} \Phi^D \Delta^D (\Gamma^D \mathbf{S}^F + \Lambda^F) \Phi^F \Psi^F,
\end{aligned}$$

where $\eta^F = \left[\frac{\partial \log p_1}{\partial \log e}, \dots, \frac{\partial \log p_F}{\partial \log e} \right]^T$ is the vector of exchange rate pass-through rates in the retail price of imported goods. In the second equation, we rewrite η^F as the product of sectoral distribution margins, Φ^F , and the sensitivity of the local-currency producer price of the imported good to exchange rates, Ψ^F .

Lastly, given the assumption on the aggregate consumption bundle and the model-based measure of inflation and pass-through, Equation (6) in the main text follows immediately:

$$\eta \equiv \frac{\partial \log \pi}{\partial \log e} = \beta^D * \eta^D + \beta^F * \eta^F. \quad (33)$$

B Appendix - Data and Empirics

B.1 Markup and Markup Elasticity Estimation

The following section provides a detailed description of the estimation of markups and markup elasticities.

Data construction I use data from the Annual Survey of Manufacturing (ENIA), administered by the Chilean National Institute of Statistics (INE), covering the years 1995 - 2015. The data is at the establishment-year level and includes approximately 30 manufacturing industries, with roughly 4,000 observations per year. For each observation, the dataset contains information on capital stock, labor, wage bill, domestic and imported materials, revenues, and electricity consumption.²² Capital stock data are unavailable after 2015. As a robustness check, we extend the sample to 2019 by constructing the capital stock using investment and depreciation data via a perpetual inventory method. Industries are defined at the three-digit CIIU Rev. 3 level (Chilean industry classification).

Observations with zero or negative values for capital, materials, revenues, or wage bill are excluded. Additionally, I drop observations with a labor share or materials share of revenue exceeding one. To remove outliers, I exclude observations in the bottom and top 5% of labor and materials shares of revenue for each industry.

A firm-level measure of capital costs is constructed as the product of capital stock and the rental rate net of depreciation. The average real interest rate for Chile during the sample period, reported in the World Bank World Development Indicators, serves as a proxy for the rental rate of capital (Raval, 2023).²³ The rental rate is combined with sectoral depreciation rates from Oberfield and Raval (2021), after creating a concordance between NAICS and CIIU classifications.

Table 2 reports the mean, the median, and the standard deviation of the key variables used in the dataset.

Measuring Markup Under the assumption of constant return to scale, as in the theoretical framework in Section 1, Autor et al. (2020) shows that markup can be measured as the ratio of firm sales to total costs:

$$\mu_{it} = \frac{\alpha_{it}^v}{S_{it}^v} = \frac{\text{Sales}_{it}}{\text{Total Cost}_{it}}, \quad (34)$$

where α_{it}^v and S_{it}^v represent the output elasticity and the factor share of input v ($S_{it}^v = \frac{\text{Expenditure on } v}{\text{Sales}_{it}}$), respectively. The second equality follows from the CRS assumption, i.e. $\alpha_{it}^v = \frac{\text{Expenditure on } v}{\text{Total Cost}_{it}}$. In mapping Equation (34) to the data, I assume that total costs are equal to the sum of wage bill, materials expenditure, and capital costs.

²² The INE applies a small amount of noise to all variables to ensure statistical privacy. For integer variables, such as labor, I use the floor of the value reported by INE.

²³ The real interest rate represents the private sector lending rate, adjusted for the domestic inflation rate as measured by the GDP deflator.

Table 2: Summary Statistics - Firms Level Data

	Mean	Median	Sd
Revenues	4844162.47	483788.12	34550569.32
Wage Bill	507305.73	101679.23	2899456.52
Capital Stock	3856994.72	133303.42	80629588.68
Materials	2130928.83	223607.36	18598449.48
Imported Materials	412995.46	0.00	7524363.83
Observations	69389		

Note: The Table reports the mean, median, and standard deviation of key firm-level variables from the ENIA manufacturing survey, covering the period 1995–2015. All values are expressed in nominal Chilean pesos. Observations with zero or negative values for capital, materials, revenues, or wage bill are excluded. Additionally, observations with a labor share or materials share of revenue exceeding one are removed. The data are trimmed at the 5% within each industry.

Table 3: Summary Statistics - Markup and Markup Elasticity

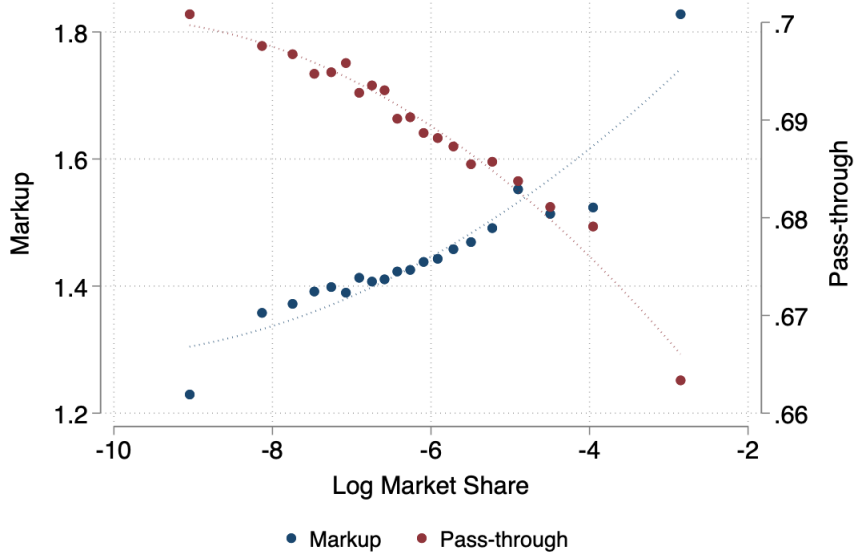
	Mean	Median	Sd	P10	P90
Markup	1.39	1.28	0.94	1.00	1.77
$\frac{\epsilon}{\sigma}$	0.34	0.29	0.11	0.19	0.51
Markup Elasticity	0.48	0.44	0.25	0.27	0.71
Pass-through	0.69	0.69	0.08	0.58	0.79
Revenue Share	0.01	0.00	0.03	0.00	0.02
Cost Share	0.01	0.00	0.03	0.00	0.02
Share Domestic Inputs	0.63	0.64	0.16	0.41	0.82
Share Imported Inputs	0.05	0.00	0.14	0.00	0.22
Observations	69389				

Note: The Table reports the mean, the median, the standard deviation, the 10th and 90th percentile of the distribution of markups, estimated $\frac{\epsilon}{\sigma}$, markup elasticities and implied pass-through rates, revenue and cost shares, domestic and imported input shares. Markups are computed as the ratio of firm sales to total costs (Autor et al., 2020). The ratio $\frac{\epsilon}{\sigma}$ is estimated using the estimated markups and Equation (37). Markup elasticities and implied pass-through rates are computed from Equation (36) using the estimated coefficients and markups. The revenue (cost) shares represent the share of each firm in the total revenues (costs) of the CIU sector. The share of domestic input is computed as the share of imported materials over total costs. The share of imported input is computed as one minus the share of domestic materials over total costs. Data are from the ENIA manufacturing survey, covering the period 1995–2015.

Table 3 shows that the estimated average and median markups are 40% and 30%, respectively, consistent with previous estimate using similar data. Furthermore, Figure 11 shows that markups increase with a firm’s market share, aligning with extensive previous literature and supporting the Marshall’s weak second law assumed in the theoretical framework presented Section 1.

Markup Elasticity and Pass-Through Measuring markups allows the estimation of the markup elasticity. To make progress, I assume that the Kimball aggregator in the distributor’s technology in Equation (7) follows the functional form proposed by Klenow and Willis (2016).

Figure 11: Relationship Market Share, Markup and Pass-through Rate



Note: The Figure plots the binscatter plot between the log market share and the markup (left axis) and the pass-through κ_{ik} (right axis) after absorbing for industry fixed effects. The market share is compute as the share of each firm in the total revenues of the CIU sector. Markups are constructed as the ratio of firm sales to total costs (Autor et al., 2020). Pass-through rates are computed under the Kimball aggregator assumption, using the estimated markups.

Thus, the inverse demand for each variety k can be written as follows:

$$\mathcal{K}'_i = \frac{\sigma}{\sigma - 1} \exp \left(\frac{1 - \left(\frac{y_{ki}}{y_i} \right)^{\frac{\epsilon}{\sigma}}}{\epsilon} \right). \quad (35)$$

This formulation depends on two parameters: the demand elasticity, $\sigma > 1$, and the demand superelasticity, ϵ . Importantly, the markup elasticity of firm k in sector i , $\Gamma_{ki} = -\frac{\partial \log \mu_{ki}}{\partial \log p_{ki}}$, takes the following function form:

$$\Gamma_{ki} = \frac{\epsilon}{\sigma} \mu_{ki}, \quad (36)$$

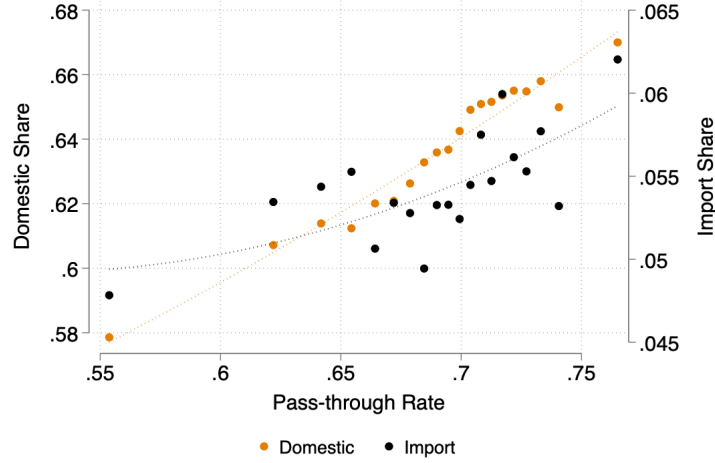
which can be computed given σ , ϵ , and firm-level markups μ_{ki} .

I follow Edmond et al. (2018) and estimate the elasticity and superelasticity of demand (σ and ϵ) to calibrate the Kimball aggregator and compute firms' markup elasticity. The Klenow and Willis (2016) specification for the Kimball aggregator in Equation (35) allows to specify the following within-industry relationship between markups and market shares:

$$\frac{1}{\mu_{ki}} + \log \left(1 - \frac{1}{\mu_{ki}} \right) = \text{constant} + \gamma \log s_{ki} + \nu_{ki} \quad (37)$$

where $\gamma = \frac{\epsilon}{\sigma}$ and s_{ki} is the market share of firm k in sector i . I estimate this specification for each sector in the economy (CIU industry), introducing firm and year fixed effects to control for unobserved productivity and/or quality differences as well as aggregate dynamics. I also exclude observations with market shares lower (larger) than the 5th (95th) percentile within each industry.

Figure 12: Correlation Pass-through - Domestic and Imported Inputs



Note: The Figure plots the relationship between firm-level pass-through rates κ_{ik} and the share of domestic (left axis) and imported inputs (right axis), after absorbing industry fixed effects. Pass-through rates are computed under the Kimball aggregator assumption, using the estimated markups. The share of domestic input is computed as the share of imported materials over total costs. The share of imported input is computed as one minus the share of domestic materials over total costs. Data are from the ENIA manufacturing survey, covering the period 1995–2015.

Table 3 reports moments of the estimated $\frac{\epsilon}{\sigma}$, the implied markup elasticities, and the pass-through rates (where the pass-through rate is $\kappa_{ki} = \frac{1}{1+\Gamma_{ki}}$). The average estimated value for $\frac{\epsilon}{\sigma}$ is 0.34, consistent with the calibrated value used in Amiti et al. (2019) and in macroeconomic studies of exchange rate pass-through more broadly.²⁴ The average implied markup elasticity is 0.48, which in turn implies a pass-through rate of 69%.

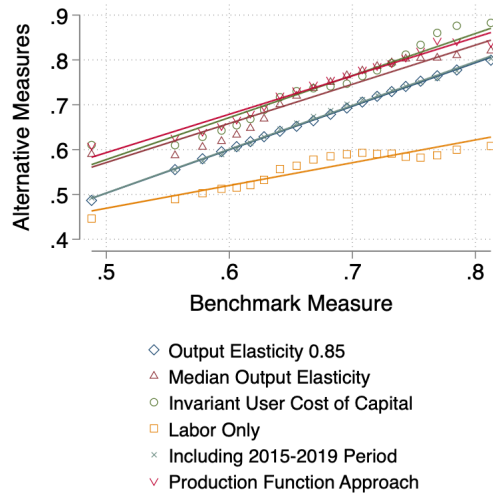
Importantly, there is substantial heterogeneity in markup elasticity and pass-through rate across firms. Given the estimated demand elasticities and the distribution of markups, the pass-through rate ranges from 0.58 to 0.8 between the 10th and 90th percentiles of the distribution. This dispersion in pass-through rates is comparable to that estimated in Amiti et al. (2019) using Belgian data.²⁵

Robustness I show that the distribution of markups used to estimate markup elasticities and pass-through rates is robust to variations in how markups are constructed. Figure 13 shows that markup values remain consistent under several key assumptions: i) using a constant output elasticity for the composite input v , α_{it}^v , equal to the sectoral median as in Raval (2023); ii) adopting a time-invariant user cost of capital of 0.14, as in De Loecker et al. (2020); iii) applying a common constant output elasticity of 0.85, as in De Loecker et al. (2020); iv) extending the sample to include data from 2015–2019, where a proxy for capital stock is

²⁴ Amiti et al. (2019) calibrates σ equal to 5 and ϵ to 1.6, implying a value of 0.32 for the ratio. A value of 0.34 is however higher than the micro-level estimates of 0.16 in Edmond et al. (2018).

²⁵ Amiti et al. (2019) estimate pass-through rates of 0.9 for small firms and 0.4 for large firms. The Chilean data used in this study replicate similar patterns, though large firms exhibit relatively higher sensitivity, with pass-through rates exceeding those found in Amiti et al. (2019).

Figure 13: Alternative Measures of Markup



Note: The Figure displays a binscatter plot between the benchmark measure of markups and the alternative measures constructed as robustness. The benchmark measure is constructed as the ratio of firm sales to total costs (Autor et al., 2020). Alternative markups are constructed: i) applying a common constant output elasticity of 0.85 (blue diamonds); ii) using a constant output elasticity for the composite input equal to the sectoral median (maroon triangles); iii) adopting a time-invariant user cost of capital of 0.14 (green circles); iv) using only labor as variable input (yellow squares); v) extending the sample to include data from 2015–2019 (gray crosses); vi) applying state-of-the-art production function estimation techniques (Levinsohn and Petrin, 2003; Akerberg et al., 2015) (red check marks). Straight lines represent a linear fit between the benchmark and the alternative markup measures. Data are from the ENIA manufacturing survey, covering the period 1995–2015.

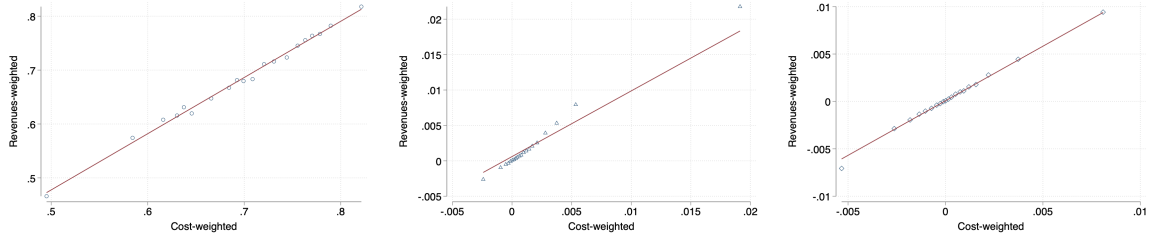
constructed.²⁶ Moreover, state-of-the-art production function estimation techniques provides very similar markup values. I construct consistent measures of inputs and outputs over time using capital, imported and domestic materials, and output deflators.²⁷ I estimate production function applying the Akerberg et al. (2015) correction to the Levinsohn and Petrin (2003) approach, using electricity consumption as a control variable and the sum of labor and material expenditure as the variable input.

The key measures of interest for the pass-through equations, the weighted average pass-through rate (Γ^D) and the covariance terms (Λ^D and Λ^F), are robust to the weighting. Figure 14 shows that the terms Γ^D , Λ^D and Λ^F constructed using cost shares exhibit a near-perfect correlation with benchmark measures constructed using revenue shares.

²⁶ In line with previous literature (e.g. Raval (2023)), markups are sensitive to the choice of the variable input v . Using only labor as variable input, markups are lower than in the benchmark case.

²⁷ For output and domestic materials, deflators are industry specific. Deflators are downloaded from the Central Bank of Chile: https://si3.bcentral.cl/siete/ES/Siete/Cuadro/CAP_CCNN/MN_CCNN76/CCNN2018_G3_A?idSerie=F033.FKF.DEF.Z.CLP.EP18.0.A for capital; https://si3.bcentral.cl/Siete/ES/Siete/Cuadro/CAP_CCNN/MN_CCNN76/CCNN1996_P3_A/CCNN1996_P3_A for output and domestic materials; https://si3.bcentral.cl/Siete/ES/Siete/Cuadro/CAP_BDP/MN_BDP42/BP6M_IND_EXPORT_A_96_11/BP6M_IND_EXPORT_A_96_11?cbFechaInicio=1996&cbFechaTermino=2011&cbFrecuencia=ANNUAL&cbCalculo=NONE&cbFechaBase= for imports and exports.

Figure 14: Revenue vs Cost Weighting



Note: The three panels plot the revenue-weighted Γ^D , Λ^D and Λ^F against the cost-weighted counterparts. The left panel plots the average pass-through, Γ^D , constructed under the Kimball aggregator assumption with markups constructed as the ratio of firm sales to total costs (Autor et al., 2020). The middle and right panels plot the covariance between pass-through rates and the aggregate share of domestic and imported inputs, respectively. The revenue (cost) shares represent the share of each firm in the total revenues (costs) of the CIIU sector. Firm-level pass-through rates and input shares are estimated using firm-level balance sheet data from the Annual Survey of Manufacturing (ENIA), administered by the Chilean National Institute of Statistics (INE), spanning the years 1995 to 2015.

B.2 Input-Output Tables

I construct the yearly input-output matrix for the Chilean economy by combining the "make" and "use" tables provided by the Central Bank of Chile (*Banco Central de Chile*) for the period 2008 to 2019.²⁸ I combine these tables to create a product-by-product input-output matrix that quantifies the extent to which each product is used in the production of other products. I opt to construct a product-by-product matrix, rather than an industry-by-industry one, in order to leverage the larger product dimension available in the make and use tables.

Following standard best practices outlined in Mahajan (2018) and Miller and Blair (2009), I construct the input-output table under the industry technology assumption. Consider the product-by-industry make matrix, V^T , and the product-by-industry use matrices for domestic and imported products, U_d and U_m , respectively. Let g^T represent the row vector of industry output, i.e. the column sum of V^T . I then construct the product-mix matrix C ,

$$C = V^T [\text{diag}(g^T)]^{-1},$$

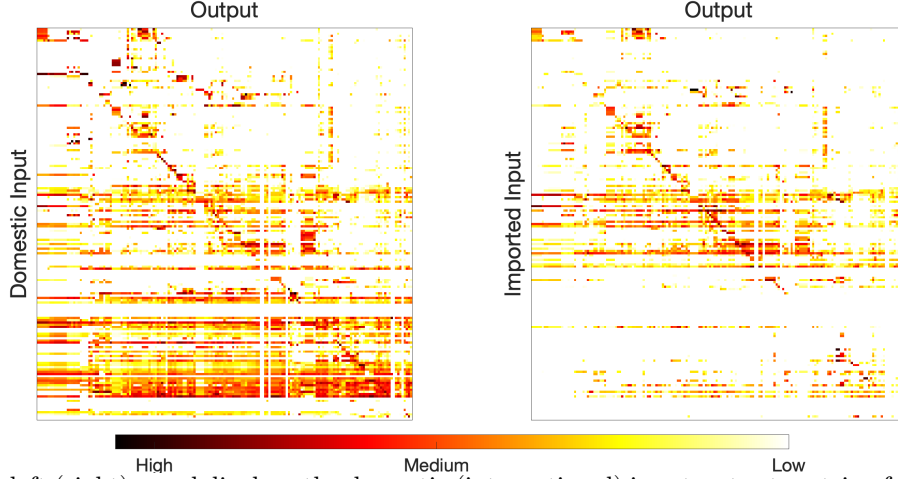
which collects the share of each product in the output of an industry. Under the industry technology assumption, each industry has its own specific production process, irrespective of its product mix.²⁹ I create the domestic and international Leontief matrices by multiplying the product-mix matrix C to the use matrices U_d and U_m :

$$S^D = U_d C^T \quad S^F = U_m C^T,$$

²⁸ Data are available at the following website: <https://www.bcentral.cl/web/banco-central/cuentas-nacionales-anales-excel>. Data before 2008 do not have the same granularity in terms of product classification.

²⁹ Compared to the more commonly used product technology assumption, the key advantage of the industry technology assumption is that it prevents the occurrence of negative elements in the input-output table.

Figure 15: Domestic and International Leontief Matrices



Note: The left (right) panel displays the domestic (international) input-output matrix of the Chilean economy, with the shares averaged over the period 2008-2012. The matrices are constructed using the make and use table under the industry technology assumption. Each row (column) represents an input (output). The color intensity indicates the extent to which one product is used as an input in the production of other products: darker (lighter) colors correspond to higher (lower) input share. Log input shares smaller than -10 are censored.

where $\mathbf{S}^D$ and $\mathbf{S}^F$ represent the domestic and international product-by-product Leontief matrices, respectively.

B.3 ERPT into Border Prices

For each import transaction, the Chilean Customs dataset includes standard information such as the importer's unique identifier (*importer*), the 8-digit HS product code (*product*), the transaction date, the country of origin (*origin*), the FOB and CIF values, and the quantity shipped. This dataset covers the period from 2007 to 2019. I compute import prices as unit values by dividing the shipment value by the quantity shipped. I aggregate the data at the importer-product-origin level at quarterly frequency. Transactions with missing information, such as missing quantity, value, or weight, are excluded from the dataset. Panel A of Table 5 provides summary statistics of the main variables while Panel B reports information on industry and origin composition of the data. For the estimation of exchange rate pass-through rates, I use bilateral exchange rates between the Chilean peso and the currency of the origin country at quarterly frequency. These exchange rates are sourced at a quarterly frequency from Bloomberg/Datastream.

B.4 Empirical CPI Sensitivity to Exchange Rates

Following [Goldberg and Campa \(2010\)](#) and [Burstein and Gopinath \(2014\)](#), I estimate the aggregate CPI sensitivity for the period 1990-2019 at the quarterly level using the following specification:

$$\Delta \log CPI_t = \sum_{\tau=0}^8 \beta_{\tau} \Delta \log e_{t-\tau} + \sum_{\tau=0}^8 \gamma_{\tau} X_{t-\tau} + \varepsilon_t, \quad (38)$$

where CPI is the Chilean consumer price index at the quarterly frequency, e is the trade-weighted nominal exchange rate between the Chilean peso and the exporting country's currency, and X is a vector of controls that includes i) the inflation rate in the exporting country as proxy for trading partners' costs (Campa and Goldberg, 2005; Burstein and Gopinath, 2014) and ii) Chile's real GDP growth rate.³⁰ I include up to eight lags to control for gradual adjustments and auto-correlation in inflation, exchange rates, and controls. Data are sourced from IMF and Datastream.

The estimated contemporaneous CPI sensitivity from Equation (38) is 8.2%, in line with estimates in the literature (Goldberg and Campa, 2010). Table 7 shows that the coefficient is robust to different specifications. The coefficient remains robust to including six or ten lags (Columns (2) and (3), respectively). Column (4) replaces with the trade-weighted exchange rate with the USD-Peso bilateral exchange rate as the majority of Chilean imports are denominated in US dollar (Gopinath et al., 2020). In this case, the coefficient is slightly lower, but still the range of previous estimates.

³⁰ Trading partners' costs are proxied by the difference between the nominal and the real effective exchange rate sourced from the IMF as in Campa and Goldberg (2005).

Table 4: Distribution Margin - Summary Statistics

Panel A:	Mean	Median	St Dev
Intermediate - Domestic	0.122	0.047	0.195
Intermediate - Imported	0.133	0.043	0.203
Final - Domestic	0.244	0.199	0.200
Final - Imported	0.384	0.398	0.172

Panel B:	Intermediate Goods		Final Goods	
	Domestic	Imported	Domestic	Imported
Farming	0.197	0.112	0.112	0.341
Fishing and Forestry	0.0146	0.0163	0.0163	0.127
Oil, Coal and Gas Extraction	0.0641	0.0212	0.0212	.
Mining	0.0252	0.0518	0.0518	0.0724
Food, Beverages and Tobacco	0.210	0.176	0.176	0.421
Textile and Apparel	0.174	0.278	0.278	0.538
Wood, Paper and Printing	0.139	0.136	0.136	0.332
Petroleum and Chemical Products	0.219	0.219	0.219	0.441
Plastic Rubber and Construction	0.0919	0.171	0.171	0.469
Fabricated Metal Products	0.0882	0.132	0.132	0.403
Machinery and Equipment	0.116	0.255	0.255	0.405
Motor Vehicles	0.0733	0.418	0.418	0.379
Furniture	0.114	0.167	0.167	0.433
Utilities	0.118	0.120	0.120	.
Construction	0.0122	0.0476	0.0476	.
Wholesale and Retail Trade	0.0293	0.000522	0.000522	0
Transportation	0.0425	0.174	0.174	0
Health Care and Education	0.0986	0.0833	0.0833	.
Accommodation and Recreation	0.0913	0.0646	0.0646	.
Professional Services	0.0636	0.0562	0.0562	0.163
Communication	0.0991	0.0860	0.0860	0
Other Products or Services	0.0983	0.0904	0.0904	0.727

Note: Panel A reports the mean, the median, and the standard deviation of the distribution margin across all products with non-zero use share. Panel B reports the average distribution margin for each industry (2-digit CIIU classification). I distinguish across products depending on their use, final vs intermediate use, and on their origin, imported vs domestically produced. The distribution margin is computed according Equation (18) using the input-output tables of Chile from 2008 to 2019.

Table 5: Summary Statistics

Panel A: Summary Statistics	Whole Sample				
	Mean	Median	StD	p5	p95
Importers	41,186	.	.	.	.
Products	7,518	.	.	.	.
Origin Countries	168	.	.	.	.
Products per importer	10.66	3	27.28	1	43
Origins per importer	2.227	1	2.931	1	7
Unit value (USD/quantity)	1,732.7	21.35	76,930.6	0.934	1,569.2
% Δ log unit value	0.446	0.417	0.690	-116.6	118.1
Transaction value (USD)	130,817.5	7,214.3	2,659,917.9	239.5	286,991.7
Observations (N)	3,044,931	.	.	.	.
Panel B: Breakdown by Industry and Origin		Numbers of Transactions (%)		Import Value (%)	
Industry (SITC):					
Food & Animals		3.871		8.238	
Beverages, Tobacco		0.291		0.613	
Crude Materials		1.589		2.392	
Mineral fuels		0.503		24.34	
Animal & Vegetable Oils		0.192		0.524	
Chemicals		11.55		13.23	
Manufactured Goods		18.57		9.466	
Machinery		36.52		33.02	
Mix Manufacturing		26.91		8.160	
Country:					
China		14.02		6.208	
USA		25.01		30.00	
EU15		25.41		17.41	
Other Americas		18.42		25.03	
Others		17.14		21.35	

Note: Panel A presents summary statistics for the cleaned sample of import transactions from Chilean Customs data covering the period 2007-2019. Transaction values and unit values are reported in USD. Panel B provides a breakdown of the cleaned universe of import transactions by industry (2-digit SITC level) and country of origin. The breakdown is computed in terms of: i) the number of transactions and ii) import values.

Table 6: Exchange Rate Pass-Through on Import Prices at the Border

	(1)	(2)	(3)	(4)
$\Delta \log e_{ot}$	0.66	0.67	0.59	0.68
	(0.02)	(0.02)	(0.02)	(0.02)
Importer-Product-Origin & Time FEs	Yes	No	Yes	No
Origin-Product & Importer-Time FEs	No	Yes	No	No
Controls	Yes	Yes	No	Yes
N	1977649	2098355	3141787	2132417

Note: The Table reports the estimated coefficients from the specifications based on Equation (22) using the cleaned sample of import transactions from Chilean Customs data covering the period 2007-2019. In all specifications the dependent variable is the quarterly log change in the price of imports at the firm-HS8-origin level. In all specification, eight lags of the log change in exchange rate are also included but not reported. In all columns but Column (3), the contemporaneous and eight lags of the origin country's inflation rate and GDP growth are included, but not reported. Columns (1) and (3) include origin-product-importer and time fixed effects, while Column (2) includes importer-time and origin-product fixed effects. Column (4) does not include any fixed effect. Standard errors clustered at origin-HS8 level.

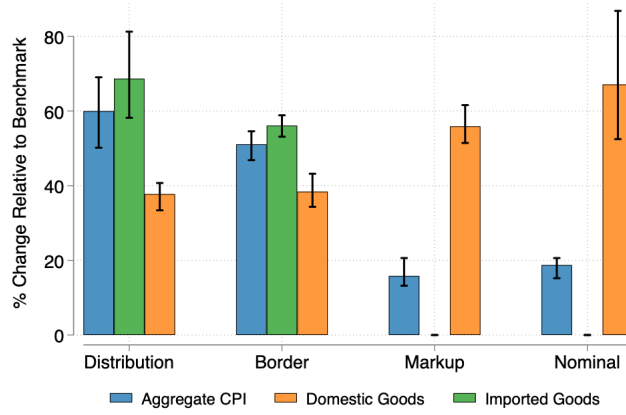
Table 7: CPI Sensitivity to Exchange Rates

	(1)	(2)	(3)	(4)
	Main	6 Lags	10 Lags	USD-CLP
$\Delta \log(\text{Weighted ER})$	0.082 (0.021)	0.079 (0.019)	0.075 (0.021)	
$\Delta \log(\text{US-CLP ER})$				0.064 (0.028)
N	120	120	120	120

Note: The Table reports the contemporaneous coefficient of $\Delta \log e_t$ from Equation (23). All columns use the trade-weighted nominal exchange rate between the Chilean peso and the exporting country's currency, except for Column (4), which uses the USD-Peso bilateral exchange rate. Controls include Chile's real GDP growth rate and proxies for trading partners' costs. Coefficients for additional variables and their lags are omitted for brevity. Newey-West standard errors are used to account for autocorrelation. Inflation and exchange rate data are sourced from IMF and Datastream, respectively.

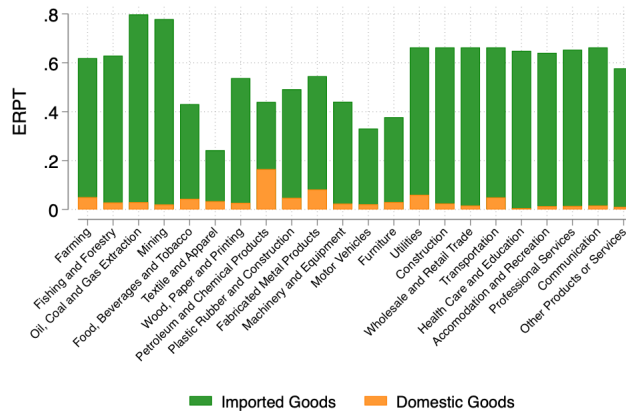
C Appendix - Additional Results

Figure 16: Individual Contribution Relative to Benchmark



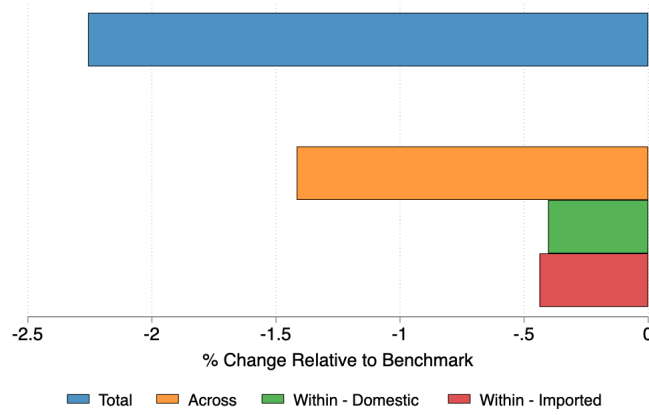
Note: The Figure depicts the percentage change in sensitivity relative to the benchmark economy when excluding one element - distribution costs, variable markups, nominal rigidities, or border price responses - at a time. Blue bars depict the percentage change in CPI sensitivity relative to the benchmark economy. Orange and green bars show the percentage change in the sensitivity of domestic and imported final goods, respectively, relative to the benchmark economy. The bands represent the range of estimated CPI sensitivities for the years 2008 - 2019.

Figure 17: Sectoral Heterogeneity



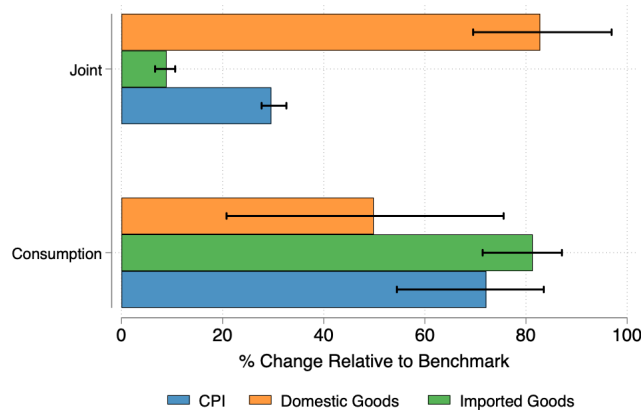
Note: The Figure presents the average sensitivity of final goods across 2-digit CIUU industries, as computed in the fully calibrated benchmark economy. For each industry, the bars represent the average sensitivity of domestically produced final goods (orange) and imported final goods (green).

Figure 18: Within Sector Heterogeneity



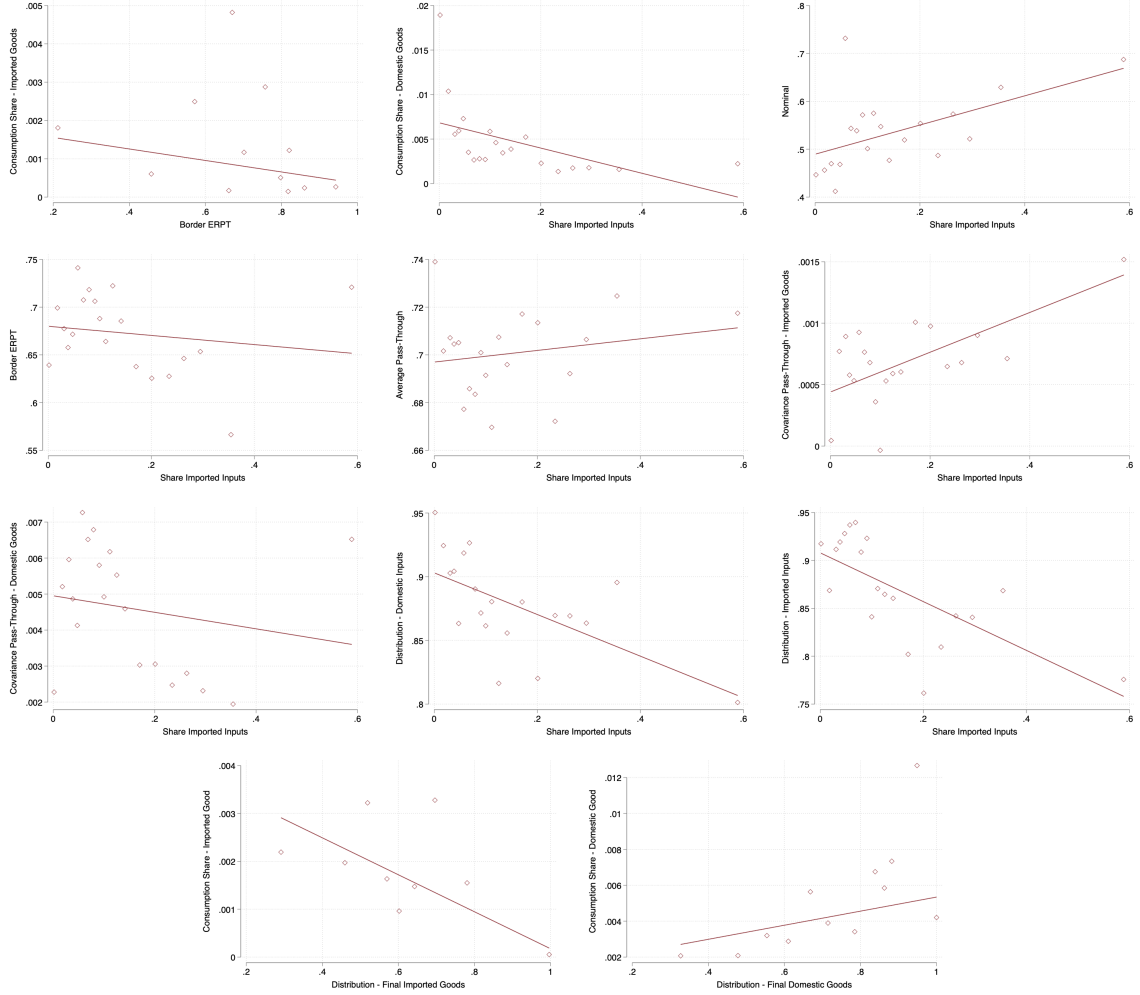
Note: The Figure presents the percentage change in the sensitivity of domestic final goods relative to the benchmark economy when omitting heterogeneity in markups. The blue bar shows the total effect, i.e. when abstracting from both heterogeneity within and across products (Γ^D equal to the average across products and $\Lambda^F = \Lambda^D = 0$, respectively). The orange bar shows the effect of abstracting from heterogeneity across products only. The green and red bars show the effect of abstracting from within-product heterogeneity in domestic and imported inputs, respectively. Numbers are averaged over the years 2008-2019.

Figure 19: Heterogeneity in Consumption Shares



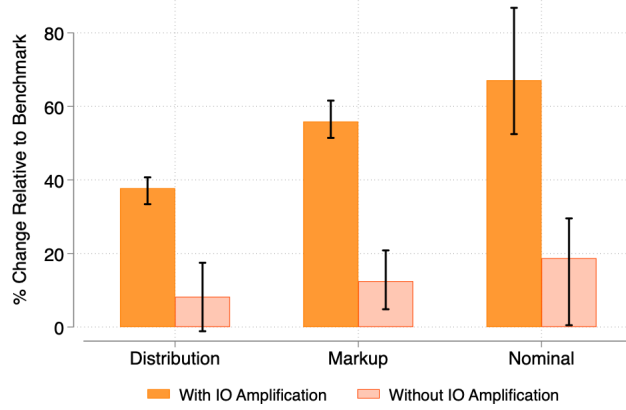
Note: The Figure presents the percentage change in CPI sensitivity relative to the benchmark economy when omitting heterogeneity in consumption shares. The "Consumption" scenario omits the heterogeneity in consumption shares only. Similarly, the "Joint" scenario consider the heterogeneity in consumption share and the heterogeneity in all wedges. Orange and green bars show the percentage change in the sensitivity of domestic and imported final goods, respectively, relative to the benchmark economy. The bands represent the range of estimated CPI sensitivities for the years 2008-2019.

Figure 20: Correlation in Primitives



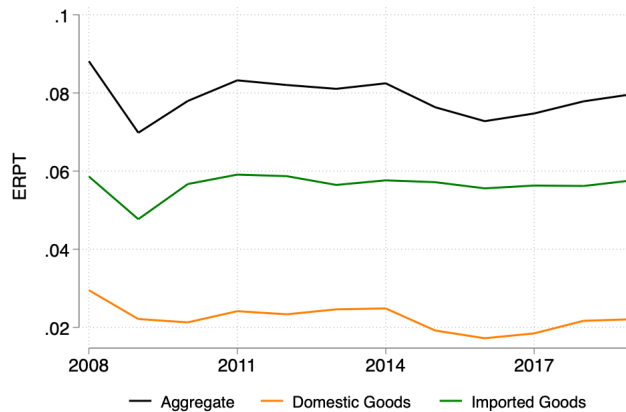
Note: Each panel depicts a binscatter plot between two primitive characteristics of the economy, with a solid line representing a linear fit. Correlation coefficients corresponding to these relationships are reported in Table 1. Top row: The left panel shows the relationship between border price sensitivity and the consumption share of imported final goods. The middle and right panels plot the aggregate share of imported inputs used in production against the consumption share of domestic goods in the final consumption basket and the frequency of price adjustment, respectively. Second row: The left, middle, and right panels display the relationship between the aggregate share of imported inputs used in production and the average border price sensitivity of imported inputs, the average pass-through rate, and the covariance between pass-through rates and domestic goods, respectively. Third row: The left, middle, and right panels show the relationship between the aggregate share of imported inputs used in production and the covariance between pass-through rates and imported goods, the distribution margins of domestic inputs, and the distribution margins of imported inputs, respectively. Bottom row: The left and right panels illustrate the relationship between the distribution margin and the consumption share of domestic and imported goods in the final consumption basket, respectively.

Figure 21: Effects of Input-Output Linkages



Note: The Figure depicts the percentage change in CPI sensitivity relative to the benchmark economy when excluding one element - distribution costs, variable markups, nominal rigidities - at a time. The darker bars depicts the effect accounting for the amplification effects of domestic input-output linkages. In contrast, the lighter bars represents the effect when the amplification from domestic input-output linkages is omitted. The bands represent the range of estimated CPI sensitivities for the years 2008 - 2019.

Figure 22: Stability Over Time



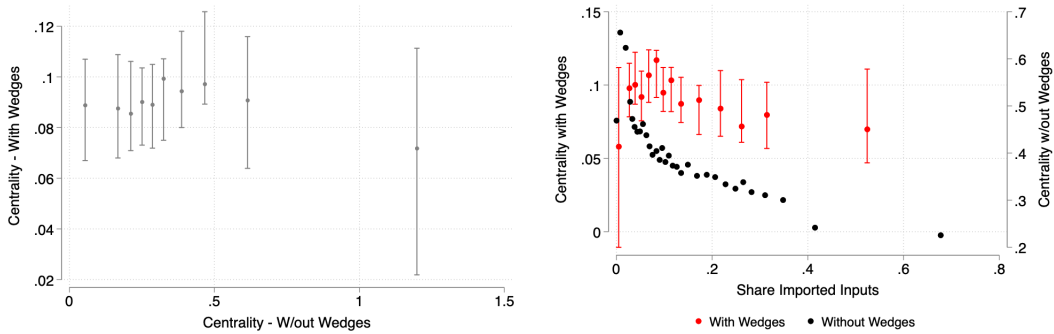
Note: The Figure illustrates the sensitivity in the fully calibrated benchmark economy over time, spanning from 2008 to 2019. The black line represents the sensitivity of the CPI, while the orange and green lines represent the sensitivity of domestic and imported final consumption goods, respectively.

Table 8: Alternative Calibrations

	Aggregate	Domestic Goods	Imported Goods
Benchmark	0.079 (0.0050)	0.022 (0.0033)	0.056 (0.0030)
Complete Border ERPT Service	0.084 (0.0049)	0.025 (0.0030)	0.059 (0.0031)
Zero Border ERPT Service	0.069 (0.0060)	0.017 (0.0043)	0.052 (0.0030)
Cost-weighted Markup Measures	0.079 (0.0051)	0.023 (0.0034)	0.056 (0.0030)
Rescaled Nominal Rigidities	0.082 (0.0051)	0.025 (0.0033)	0.056 (0.0030)

Note: The Table presents the sensitivity of the CPI, domestic, and imported final goods in the benchmark economy (top row) and across various alternative calibrations (remaining rows). The second and third rows calculate sensitivities by setting the sensitivity of import prices for service products at the border to one and zero, respectively, instead of using the average sensitivity across manufacturing products. The fourth row computes sensitivities using cost shares rather than revenue shares to weight the average pass-through and covariance terms arising from variable markups. The final row calculates sensitivities by rescaling the frequencies of price adjustment to align with the average Calvo probability reported in [Aruoba et al. \(2022\)](#). Standard deviations of sensitivities over the 2008–2019 period are reported in parentheses.

Figure 23: Product Centrality and Imported Inputs



Note: The left panel illustrates the relationship between the centrality of a product in the input-output network when the edges are constructed based on input-output tables that include domestic wedges (y-axis) versus those that omit domestic wedges (x-axis). The right panel depicts the relationship between a product's centrality in the domestic production network and the share of imported inputs used in its production. Centrality is measured using the centrality measure, with edges weighted by input-output linkages that either include domestic wedges (red dots) or omit domestic wedges (black dots). Centrality of a product is measured as the average between its indegree and outdegree centrality of node in the input-output network. Both panels use data spanning the years 2008 to 2019. Both panels use binscatters constructed while absorbing product and year fixed effects. The figures include 95% confidence intervals.

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